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A New Framework for Lagrangian Hydrodynamics with the Laguerre Diagram

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Abstract

We present `LagEunha`, a fully Lagrangian hydrodynamics code built on the Laguerre (power) diagram—a weighted generalization of the Voronoi tessellation. Because every cell boundary moves exactly at the local material velocity, the advection flux $\nabla \cdot (\phi \boldsymbol{u})$ that dominates the truncation error in Eulerian and Arbitrary-Lagrangian–Eulerian (ALE) methods is absorbed *analytically* into the material derivative, leaving only the surface pressure–work integral on the right-hand side of the Euler equations. Mass, momentum, and energy conservation therefore hold *by construction* at every time step, without recourse to advection limiters, remapping diffusion, or Galilean-invariance corrections.

The scheme used in the tests is the geometric-face scheme (GFS). Every Laguerre weight is zero, so each face lies on the Voronoi mid-plane. On an ordinary face the pressure is the MUSCL–HLLC pressure and the face moves at the mean of the two particle velocities. On an extreme face, where the pressure ratio exceeds 100 or the lower pressure falls below 10^{-3}, the pressure and the normal velocity are taken from the Harten–Lax–van Leer average of the conserved state. GFS is integrated on the Shu–Osher problem, a two-shock collision, a Kelvin–Helmholtz shear layer, a Rayleigh–Taylor layer, and Mach 3 flow past a solid cylinder. The geometric face velocity keeps the Rayleigh–Taylor energy conserved while the dense spikes reach the lower boundary. The cylinder wall is a reflecting Riemann problem, and no particle enters the body.

Unified Astronomy Thesaurus concepts: Galaxy interactions (600), Galaxy encounters (592), Galaxy dark matter halos (1880), Large-scale structure of the universe (902), Cosmic web (330), Galaxy kinematics (602), Dark matter (353), N-body simulations (1083)

1. Introduction

Numerical hydrodynamics lies at the heart of modern astrophysical simulation. Three broad families of methods are in widespread use: (i) *Eulerian* codes such as `Ramses` and ENZO, which solve the Euler equations on a fixed or adaptively refined mesh; (ii) *Lagrangian* particle methods such as Smoothed Particle Hydrodynamics (SPH; Lucy 1977; Gingold & Monaghan 1977) implemented in GADGET and GIZMO; and (iii) *Arbitrary-Lagrangian–Eulerian* (ALE) or moving-mesh methods, exemplified by the Voronoi-based code AREPO. Each approach carries intrinsic trade-offs. Eulerian schemes require advection fluxes across every cell face at every time step. Computing these fluxes demands high-order reconstruction (WENO, TVD, PPM), yet the truncation error of the advection operator—proportional to the mesh velocity relative to the flow—remains the dominant source of numerical diffusion. ALE methods alleviate this by letting the mesh partially follow the flow, but a remapping (or mesh-regularization) step is still needed to prevent cell distortion, and this remapping reintroduces advective diffusion. SPH avoids a mesh altogether, but its kernel-based density estimate introduces smoothing errors that degrade contact discontinuities and suppress fluid instabilities.

The key observation motivating this work is that *all* of these difficulties stem from the same root cause: the need to evaluate or approximate the advection flux $\nabla \cdot (\phi \boldsymbol{u})$. In a *fully Lagrangian* framework, where every cell boundary moves at the exact material velocity, the advection flux is absorbed analytically into the Lagrangian (material) derivative. The conserved form of the Euler equations (eq. 11) then reduces to the material-derivative form (eq. 14):

$$\mathcal{D}_t^{\Omega}(\mathsf{U}) = -\oint_{\partial\Omega} \mathsf{G}_{\rm ext} \cdot d\boldsymbol{S} + \int_{\Omega} \mathsf{f}_s \, dV,$$

where Ω is a Lagrangian cell that moves with the flow. Mass conservation becomes an identity ($\mathcal{D}_t^{\Omega}(\rho) = 0$), momentum and energy are updated solely through the surface pressure–work integral, and no advection limiter, remapping operator, or Galilean-invariance correction is required. We term this property *self-consistent conservation*: the conservation laws

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hold by construction at every stage of the time integration, as a direct consequence of the Lagrangian cell definition.

We present LagEunha (**Lag**uerre tessellation **E**ngine for the **Un**iverse with Numerical **H**ydrodynamics **A**lgorithms; "Eunha" means *galaxy* in Korean), a new hydrodynamics code that realizes this fully Lagrangian ideal. LagEunha tessellates the computational domain with the *Laguerre* (power) *diagram*—a weighted generalization of the Voronoi diagram in which each generator carries an adjustable power distance ω_p. GFS sets every power distance to zero. The mesh is then the Voronoi diagram of the particles, and each face moves with the particles. No remapping step is applied.

The advantages of this fully Lagrangian, self-consistent framework over the three conventional paradigms are summarized as follows.

1. **Versus Eulerian methods.**— The advection flux is eliminated entirely, removing the dominant source of numerical diffusion. No reconstruction limiter (WENO, TVD) is needed on the left-hand side of the equations, and the CFL condition involves only the sound speed rather than the sum of the flow and sound speeds. Spatial resolution adapts automatically with the mass distribution, making AMR-style grid refinement unnecessary.
2. **Versus ALE / moving-mesh methods.** No remapping step is applied. The cell faces are the Voronoi mid-planes of the particles, and the face velocity used for the $P\,dV$ work is the velocity of that mid-plane except on an extreme face.
3. **Versus SPH.**— Cell-averaged densities are exact by construction ($\rho_p = m_p/\Omega_p$), whereas SPH estimates density through a kernel sum that introduces smoothing errors. Contact discontinuities and fluid instabilities are therefore resolved sharply from the outset. The face-based pressure evaluation provides a natural framework for Riemann solvers and high-order face interpolations that are difficult to formulate in the kernel-based SPH context.

The hydrodynamics scheme in this paper is the geometric-face scheme (GFS). GFS uses a fourth-order Runge–Kutta integrator. It is tested on the Shu–Osher problem, a two-shock collision, a Kelvin–Helmholtz layer, a Rayleigh–Taylor layer, and Mach 3 flow past a cylinder. Earlier face algorithms, including signal-velocity artificial viscosity, $M(n,m)$ averaging, and the Cullen–Dehnen switch, are recorded in the appendices and are not part of GFS.

Throughout this paper the gas is adiabatic, with $P = (\gamma - 1)u$. Here γ is the adiabatic index and u is the internal energy per unit mass. Scalar and vector variables are set in italic type, and tensors and the gradient are set in bold. Section 2 derives the Lagrangian conservation laws. Section 3 defines the cell and the GFS face velocity. Section 4 discretizes momentum and energy. The later sections present the parallel layout, the GFS tests, and the discussion.

2. Full Lagrangian Perspectives of Hydrodynamics

The finite mass method (FMM) is adopted in this paper. The method assumes that a gas element follows the flow of fluid and the interactions between neighboring elements are made on the contact surfaces between the elements.

2.1. General Expression of Conservation Laws

The fluid equation for a general function, ϕ, is expressed as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi u) = \nabla \cdot (a_\phi \nabla \phi) + S^\phi \tag{1}$$

where the first term in the right-hand side of the equation is the diffusion term and the second one is the source term. The left-hand side of the equation are about the conserved quantities while the right-hand side is for the influx or the source term. Unless there is any influx or source term, the integral form of it states as

$$\int_\Omega \frac{\partial \phi}{\partial t} dV + \int_\Omega \nabla \cdot (\phi u) dV = 0 \tag{2}$$

or according to the Gauss theorem (or the divergence rule) it reads

$$\int_\Omega \frac{\partial \phi}{\partial t} dV + \oint_{\partial\Omega} \phi u \cdot dS = 0 \tag{3}$$

where dS is the surface of the volume element dV and has the outward direction. This equation implies that the increase of the fluid quantity ϕ in the volume (the first term in the left-hand side of the equation) is equal to the mass flux coming through the volume surface.

2.2. Lagrangian Alternatives

However, the left-hand side of equation (1) without the source and diffusion terms can be rewritten as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi u) = \frac{d\phi}{dt} + \phi \nabla \cdot u \tag{4}$$

where we have used the chain rule for the total derivative as $d/dt \equiv \partial/\partial t + u \cdot \nabla$. Now for a conserved quantity, we multiply it with the Lagrangian material volume (δV) of a gas cell as

$$\frac{d\phi}{dt} \delta V + \phi (\nabla \cdot u) \delta V = 0. \tag{5}$$

If we assume that the gas cell moves along with the flow (or more specifically, the every point inside the gas cell follows the flow). Then, the change of the cell is equal to the velocity divergence as

$$(\nabla \cdot u) \delta V = \frac{d}{dt} (\delta V). \tag{6}$$

Finally we can get the Lagrangian version of the conservation rule as

$$\frac{d}{dt} (\phi \delta V) = 0, \tag{7}$$

or in the integral form,

$$\frac{d}{dt}\int \phi dV = 0. \tag{8}$$

which plays a very important role in transforming the hydrodynamic equations throughout this paper.

We introduce a canonical operator,

$$\mathcal{D}_t^{\Omega}(\phi) \equiv \frac{d}{dt}\int_{\Omega} \phi dV \tag{9}$$

where we call $\mathcal{D}_t^{\Omega}$ the conservative (or material) operator for the Lagrangian moving volume Ω, hereafter. This equation implies that a certain quantity of a gas cell (ϕ) is conserved as it moves along the gas flow unless there is any external flux or source terms. This conservation rule holds for any gas conserved quantities such as the density, velocity, energy and etc.

2.3. Connection between Eulerian and Lagrangian Expressions

According to the finite volume method, the Eulerian expression in the conserved form of ϕ can easily change to the Lagrangian expression as

$$\int \left(\frac{\partial \phi}{\partial t} + \nabla(\phi u)\right) dV \rightarrow \mathcal{D}_t(\phi) \tag{10}$$

where the diffusion term in the Eulerian scheme is not shown anymore in the Lagrangian version. So, this simplicity is a key advantage to developing a new Lagrangian method for the hydrodynamics.

The conserved form of Euler equations are usually expressed as

$$\frac{\partial \mathsf{U}}{\partial t} + \nabla \cdot \mathsf{F} = -\nabla \cdot \mathsf{G}_{\rm ext} + \mathsf{f}_{\rm s} \tag{11}$$

where

$$\mathsf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho e \end{pmatrix}, \quad \mathsf{F} = u\mathsf{U} = \begin{pmatrix} \rho u \\ \rho uu \\ \rho eu \end{pmatrix}, \quad \mathsf{G}_{\rm ext} = \begin{pmatrix} 0 \\ \boldsymbol{T} \\ \boldsymbol{T}\cdot u \end{pmatrix}. \tag{12}$$

Or in the volume-integral form,

$$\int_{\Sigma} \frac{\partial \mathsf{U}}{\partial t} dV + \oint_{\partial\Sigma} \mathsf{F}\cdot dS = -\oint_{\partial\Sigma} \mathsf{G}_{\rm ext}\cdot dS + \int_{\Sigma} \mathsf{f}_s dV \tag{13}$$

where Σ is a static or moving mesh cell, and $\partial\Sigma$ is the surface of the cell. The total stress tensor ($\boldsymbol{T}$) involves contributions from the magnetic field ($\boldsymbol{B}$), external surface pressure ($\mathcal{P}$), and surface stress ($\boldsymbol{\tau}$) being expressed as $\boldsymbol{T} = (\mathcal{P} + B^2/2)\boldsymbol{I} + \boldsymbol{\tau} - B^{\mathsf{T}}B$. To solve this hyperbolic differential equations in the Eulerian scheme, many methods are developed in the schemes of FEM, FDM, FVM, and hybrids and tested up to a possible order of magnitude.

It is worthwhile to discuss the advection term of $\nabla\cdot\mathsf{F}$ in equation (11) in the Eulerian scheme. This term deals with the flux of a quantity through the surface of an Eulerian volume. To get a flux on the face of a cell, however, we should interpolate the face-centered value with discrete neighboring cell-centered values, which introduces numerical errors due to the diffusion caused by the second or higher-order derivatives. So, high-orders of derivatives are needed to suppress the numerical diffusion like the WENO/TVD/upwind schemes.

However, in the pure Lagrangian counterparts, we may have a different form as

$$\mathcal{D}_t^{\Omega}(\mathsf{U}) \equiv \frac{d}{dt}\left(\int_{\Omega} \mathsf{U} dV\right) \tag{14}$$

$$= -\oint_{\partial\Omega} \mathsf{G}_{\rm ext}\cdot dS + \int_{\Omega} \mathsf{f}_s dV, \tag{15}$$

where Ω is the Lagrangian (material) cell and $\partial\Omega$ is its surface. It simplifies the left-hand side of equation (13) by plugging in the advection term into the Lagrangian derivative, (d/dt), of the volume integration. This may help to suppress spurious numerical diffusions caused by the high-order approximation made for calculation of the advection term of $\nabla\cdot\mathsf{F}$. Also it provides the conservation of laws self-consistently along the evolution of a material cell.

Therefore, we highlight our new method how to represent the Lagrangian gas cell and calculate the corresponding the face-averaged value of $\mathsf{G}_{\rm ext}$ on the surface.

2.4. Mass Conservation

In this case, we may set $\phi = 1$ and, then, we get the mass conservation, which means that if a gas cell moves along the fluid flow, the mass conservation is satisfied consequently as

$$\mathcal{D}_t^{\Omega}(\rho) = 0. \tag{16}$$

2.5. Momentum Equation

In this case, we can set $\phi = u$ because each vector component independently satisfies the conservation rule. Then, the Euler equation of the momentum can be transformed into the Lagrangian version as

$$F(\Omega) \equiv \mathcal{D}_t^{\Omega}(\rho u) = \int_{\Omega} (-\nabla\cdot\boldsymbol{\sigma} + \mathsf{f}_{\rm s}) dV \tag{17}$$

$$= -\int_{\partial\Omega} \boldsymbol{\sigma} dS + \int_{\Omega} \mathsf{f}_{\rm s} dV \tag{18}$$

where $F(\Omega)$ is the force experienced by the Lagrangian material volume Ω and the magnetic contributions are neglected. And $\boldsymbol{\sigma}$ is the stress tensor being decomposed into

$$\boldsymbol{\sigma} \equiv \boldsymbol{\tau} + \mathcal{P}\boldsymbol{I}, \tag{19}$$

where $\mathcal{P}$ is the pressure, $\boldsymbol{I}$ is the identity tensor and $\boldsymbol{\tau}$ is called the viscous stress tensor expressed with velocity divergence and gradient as

$$\boldsymbol{\tau} \equiv -\nu\rho\left(\boldsymbol{\nabla u} + \boldsymbol{\nabla u}^{\mathsf{T}} - \frac{2}{3}\boldsymbol{I}\nabla\cdot u\right) \tag{20}$$

where $\boldsymbol{\nabla u}$ is a gradient tensor.

When the external force is absent, the momentum equation can be shown as

$$F(\Omega) = -\int_{\Omega} \nabla\cdot(\mathcal{P}\boldsymbol{I} + \boldsymbol{\tau})dV \tag{21}$$

$$= -\oint_{\partial\Omega} \mathcal{P}dS + \oint_{\partial\Omega} \boldsymbol{\tau}\cdot dS \tag{22}$$

where we used the identity relation, $\boldsymbol{I}\cdot dS = dS$.

2.6. Energy Equation

The energy conservation can be expressed as

$$\mathcal{D}_t^{\Omega}(\rho e) = -\frac{d}{dt}W \tag{23}$$

where W is the work done to the neighboring cells and can be calculated as

$$\mathcal{D}_t^{\Omega}(\rho e) = \int_{\Omega} \nabla\cdot(\mathcal{P}u + \boldsymbol{\tau}\cdot u)dV \tag{24}$$

$$= \oint_{\partial\Omega} (\mathcal{P}u + \boldsymbol{\tau}\cdot u)\cdot dS. \tag{25}$$

On the other hand, the total specific energy (e) inside the Lagrangian material volume is the sum of specific kinetic (k) and internal (u) energies.

2.7. Time Evolution of the Magnetic Field

The time evolution of the non-ideal magnetic field can be written as

$$\frac{\partial B}{\partial t} = \nabla\times(u\times B) + \eta\nabla^2 B \tag{26}$$

where the second term in the right-hand side is the diffusion term. This equation can be transformed to

$$\frac{dB}{dt} + B(\nabla\cdot u) = \nabla\cdot(B\otimes u) + \eta\nabla^2 B \tag{27}$$

where we have used the relation $\nabla\cdot B = 0$. If we apply the volume integration to the equation, then we get the Lagrangian formulation as

$$\mathcal{D}_t^{\Omega}(B) \equiv \frac{d}{dt}\left(\int_{\Omega} BdV\right) = \left(\oint_{\partial\Omega} (B\otimes u + \eta\nabla B)\cdot dS\right)$$
$$= \oint_{\partial\Omega} B(u\cdot dS) + \eta\oint_{\partial\Omega} \nabla B\cdot dS \tag{28}$$

which means that the change of the magnetic field in the material volume is the sum of incoming magnetic flux (first term) and the magnetic diffusion (second term).

It is noteworthy that equation (28) is a conserved form of B since the surface vector (dS) is anti-symmetric between neighboring pair when B and u are defined on $\partial\Omega$.

2.8. Cell-Averaged Stress Tensor

For the cell-averaged approximations, the velocity gradient and velocity divergence at the position of particle i can be obtained using the divergence theorem as

$$\langle\boldsymbol{\nabla u}\rangle = \frac{1}{\Omega}\int_{\Omega} \boldsymbol{\nabla u}dV \tag{29}$$

$$= \frac{1}{\Omega}\oint_{\partial\Omega} udS \tag{30}$$

$$= \frac{1}{\Omega}\sum_{q\in\mathbb{N}(p)} u_{pq}\otimes\Delta S_{pq} \tag{31}$$

$$\tag{32}$$

where $\otimes$ is the outer product and V_i is the volume of the i'th particle cell. Likewise, we get the velocity divergence as

$$\langle\nabla\cdot u\rangle = \frac{1}{\Omega}\sum_{q\in\mathbb{N}(p)} u_{pq}\cdot\Delta S_{pq}, \tag{33}$$

Then, the face-averaged values are approximated as shown in the previous subsection.

3. Laguerre Cell Hydrodynamics

3.1. Laguerre (Power) Diagram

The Laguerre diagram or power diagram is a generalized form of the Voronoi diagram, adopting the circular radius (w_p) of each site $p \in M$ in the Euclidian space ($\mathbb{E}^{\mathrm{d}_N}$) as

$$\Omega(p) = \{\forall x \in \mathbb{E}^{\mathrm{d}_N} \mid d_{\omega}(x,p) \le d_{\omega}(x,q), \forall q \neq p\} \tag{34}$$

where $d_{\omega}^2(x,p) \equiv \|x-p\|^2 - \omega_p^2$, ω_p's are hereafter called the Laguerre distance (or power distance), and d_N is the number of dimensions. Then, $\Omega(p)$ forms a convex polytope. If $\Omega(p)\cap\Omega(q) \neq \varnothing$ for $p \neq q$, we call q a Laguerre neighbor to p. In this case, we may define the contact facet between p and q as $\partial\Omega_{pq} \equiv \Omega(p)\cap\Omega(q)$. A set of Laguerre neighbors for p is a collection of facets ($\partial\Omega_p \equiv \bigcup_q \partial\Omega_{pq}$) forming a convex polyhedron in $\mathbb{E}^3$ (or a convex polygon in $\mathbb{E}^2$). Also, we define a group of sites neighboring to p as

$$\mathbb{N}(p) = \{q \mid \Omega(p)\cap\Omega(q) \neq \varnothing, \forall q \in M - \{p\}\}. \tag{35}$$

The radical plane cuts the connecting line between p and q at the distance d_{pq}^{rad} from p being formulated as

$$d_{pq}^{\mathrm{rad}} = \frac{d_{pq}}{2}\left(1 + \frac{\omega_p^2 - \omega_q^2}{d_{pq}^2}\right), \tag{36}$$

where $d_{pq} \equiv \|q-p\|$. To put an additional constraint that the radical axis be placed in a line between p and q (or $0 < d_{pq}^{\mathrm{rad}} < d_{pq}$), equation (36) may be placed on the condition, $\|\omega_p^2 - \omega_q^2\| \le d_{pq}^2$. Since this relation should be satisfied for

all neighboring q, we finally get the upper boundary of ω_p as, $\omega_p^2 < \omega_{\rm ceil}^2 \equiv \min_q \left(\omega_q^2 + d_{pq}^2\right)$, $\forall q$. However, this cap needs the pre-determined ω_q which is also depending on ω_p. To break this circular dependence or feedback loop, therefore, we simply insert a stringent upper bound as,

$$\omega_{p,\rm ceil} \equiv \min_{q\in\mathbb{N}(p)} (d_{pq}). \tag{37}$$

In case of the Voronoi diagram ($\omega_p = \omega_q$), we simply recover the relation $d_{pq}^{\rm rad} = d_{pq}/2 \equiv d_{pq}^{\rm voro}$ as well. In Appendix J, we describe how to practically generate the Laguerre diagram.

3.2. Weights

GFS sets $\omega_p = 0$ for every particle. The Laguerre face then lies on the Voronoi mid-plane. Nonzero weights, including a pressure weight and filters that move a face only at a volume extremum, were integrated on the two-dimensional Sedov blast. The blast energy diverged within a few tens of steps. The tracer formulae are given in Appendix A.2.

3.3. A Velocity of the Laguerre Face

Let us list several notations before addressing the velocity of a Laguerre radical plane. First, we define e_{pq} as the normal vector to the Laguerre radical plane and $e_{pq} \equiv (q-p)/\|q-p\|$. The velocity difference between p and q is defined as $u_{pq} \equiv u_q - u_p$ and the velocity of the face is $u_{pq}^{\rm rad}$. From equation (36) and $r_{pq}^{\rm rad} = d_{pq}^{\rm rad} e_{pq}$, we get the velocity of the radical plane as

$$u_{pq}^{\rm rad} = \frac{1}{2}\left(1 + \frac{\omega_p^2 - \omega_q^2}{d_{pq}^2}\right) u_{pq} + \left[\frac{1}{2d_{pq}}\frac{d}{dt}\left(\omega_p^2 - \omega_q^2\right) - \frac{\omega_p^2 - \omega_q^2}{d_{pq}^2}\|u_{pq}\|\right] e_{pq} \tag{38}$$

where we used $\|u_{pq}\| = (d/dt)d_{pq}$. The velocity of the radical plane can be decomposed into several physical components as

$$u_{pq}^{\rm rad} = \frac{1}{2}\left(\frac{d_{pq}^{\rm rad}}{d_{pq}} + \frac{1}{2}\right) u_{pq} + \left[\frac{1}{d_{pq}}\left(\omega_p v_p^\omega - \omega_q v_q^\omega\right)\right] e_{pq} - \left(\frac{d_{pq}^{\rm rad}}{d_{pq}} - \frac{1}{2}\right)\|u_{pq}\| e_{pq} \tag{39}$$

where the velocity of power diatance is constrained as

$$-c_s(p) \le v_p^\omega \left(\equiv \frac{\Delta\omega_p}{\Delta t}\right) \le c_s(p). \tag{40}$$

We now analyze the contributions of each component presented in equation (39) under the non-trivial condition where the Laguerre radius of the pair, $d_{pq}^{\rm rad}$, differs from half the distance between the sites, $d_{pq}/2$. The first term on the left-hand side of equation (39) represents the direct contribution of the relative velocity between the two interacting sites. The second term quantifies the effect arising from the time evolution of the Laguerre radius itself.

The last term necessitates a detailed physical interpretation, as its contribution is contingent upon a non-zero magnitude of the relative velocity difference, i.e., $\|u_{pq}\| \neq 0$. To illustrate its origin, consider a simplified situation where the two sites are receding from one another. By transitioning to a local comoving reference frame where the two sites are instantaneously static, the observed Laguerre radii (ω's) which are constant in the laboratory frame, appear to undergo a reduction within this comoving frame. This effective shrink of ω's results in a net physical outcome where the ratio $d_{pq}^{\rm rad}/d_{pq}$ asymptotically decreases toward the value of 0.5 (as shown in eq. 36). Conversely, when the two sites are approaching each other, this ratio $d_{pq}^{\rm rad}/d_{pq}$ is observed to increase toward 0.5. This effect constitutes an apparent radial dragging induced by the relative motion of the sites. We call this fictitious influence the Laguerre Drag.

Note that, in this expression, any additional contribution normal to e_{pq} is simply ignored and the unit vector of the directional cosine (e_{pq}) is related to the surface of the Laguerre cell as $e_{pq} = dS/\|dS\|$ or $dS = dSe_{pq}$. Setting every weight to zero recovers the Voronoi mid-plane. The velocity of that plane is the mean of the two particle velocities,

$$u_{pq}^{\rm geo} = \frac{1}{2}\left(u_p + u_q\right). \tag{41}$$

Equation (41) is the velocity of the mid-point.

3.4. Geometric-face scheme

GFS sets every Laguerre weight to zero. Equation (41) is then the velocity of the Voronoi mid-plane, and the cell volume changes at that velocity.

Nonzero weights do not give a stable two-dimensional scheme. A pressure weight and several filters that shift a face only at a volume extremum were integrated on the two-dimensional Sedov blast. Each choice moved faces at the shock, or it shifted faces that the filter had left fixed. The blast energy diverged within a few tens of steps. One weight per particle cannot place every face of a two-dimensional cell. The differences $\omega_p^2 - \omega_q^2$ must close around every loop of faces. A least-squares solution therefore moves faces that were required to stay fixed.

The face pressure and the face velocity in the force and in the $P\,dV$ work are taken from the MUSCL states. Let P_L and P_R denote those states, and write $P_{\rm min} = \min(P_L, P_R)$ and $P_{\rm max} = \max(P_L, P_R)$. A face is extreme when

$$P_{\rm min} < 10^{-3} \quad \text{or} \quad P_{\rm max} > 100\, P_{\rm min}. \tag{42}$$

On an ordinary face the pressure is the HLLC pressure and the face velocity is the geometric velocity,

$$\mathcal{P}_{pq} = P^{\rm HLLC}, \qquad u_{pq}^{\rm face} = u_{pq}^{\rm geo}. \tag{43}$$

On an extreme face the HLLC star state is replaced by the Harten–Lax–van Leer average of the conserved state. For wave speeds $S_L < 0 < S_R$,

$$\mathbf{U}^* = \frac{S_R\mathbf{U}_R - S_L\mathbf{U}_L - (\mathbf{F}_R - \mathbf{F}_L)}{S_R - S_L}. \tag{44}$$

The face pressure is the equation-of-state pressure of $\mathbf{U}^*$. The normal component of the face velocity is the normal speed of that average, $s^* = (\rho v_n)^*/\rho^*$. The tangential component remains the geometric velocity,

$$u_{pq}^{\rm face} = u_{pq}^{\rm geo} - \left[(u_{pq}^{\rm geo} - s^*\hat{\mathbf{n}})\cdot\hat{\mathbf{n}}\right]\hat{\mathbf{n}}. \tag{45}$$

The Riemann contact speed on an ordinary shear face does not equal the rate of change of the Voronoi volume. A Rayleigh–Taylor calculation that assigned that contact speed to every face increased its energy without a bound. The total-energy update consistent with equations (43) and (45) is

$$\frac{dE_p}{dt} = -\sum_{q\in\mathbb{N}(p)} \mathcal{P}_{pq}\,\Delta S_{pq}\cdot u_{pq}^{\rm face}. \tag{46}$$

A stationary solid wall is imposed as a reflecting Riemann problem. For a circular cylinder of radius R centred at $\mathbf{c}$, a particle at radius $r > R$ is paired with a ghost at

$$\mathbf{x}_g = \mathbf{c} + (2R - r)\,\hat{\mathbf{r}}. \tag{47}$$

The ghost has the particle density and pressure, and its radial velocity is reversed. The HLLC solution between the particle and the ghost supplies the wall pressure. The normal velocity of the wall face is zero. A particle that steps inside the cylinder is moved to radius $2R-r$, and any remaining inward radial velocity is set to zero. The particle is not placed on the surface $r = R$. Placing many particles on that surface stacks neighbouring cells.

A one-dimensional transmissive boundary whose exterior velocity points into the domain is treated by particle insertion. When the first interior particle leaves a gap larger than $1.5\Delta x$, with $\Delta x = m/\rho_{\rm ext}$, a particle that carries the exterior state is inserted. The energy account includes the energy of each inserted particle and the $P\mathbf{v}$ work of the end faces. Without this inflow the Shu–Osher shock and the two-shock collision exhaust the driver gas, and the shock lags the reference solution.

Hereafter, we call a Laguerre site a particle when it is related to the hydrodynamics.

4. Discretizations of the Basic Conservation Rules

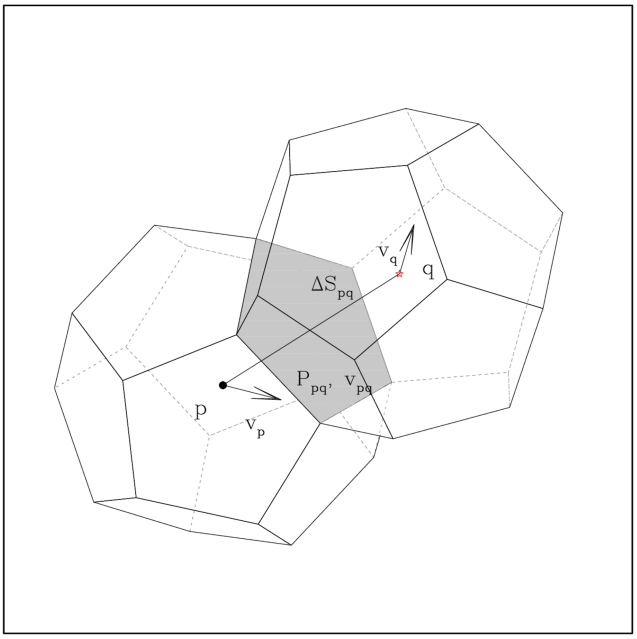

Figure 1. Hydrodynamic interactions between two Laguerre cells. The Voronoi-neighboring, p (filled circle) and q (open star symbol), share the same interacting face (shaded in gray) with the surface vector ΔS_{pq}, surface pressure P_{pq} and the surface velocity $\boldsymbol{v}_{pq}$.

We assume that the Laguerre cell is the Lagrangian control volume. Therefore, the mass conservation is self-evident because no mass flux is allowed through the Laguerre faces. Figure 1 shows an example of Laguerre cells of a pair of neighboring particles p and q. Their mutual contact surface is color coded in gray. The pressure and velocity on the surface face are labelled as P_{pq} and $\boldsymbol{v}_{pq}$ which are the basic quantities in solving the hydrodynamic equations as shown below.

4.1. Momentum Conservation

The force on a Laguerre cell of the particle p is the net force on the Laguerre-cell faces as

$$\begin{aligned}\boldsymbol{F}_p &= m_p\frac{du_p}{dt} = \mathcal{D}_t^{\Omega}(\rho_p u_p)\\ &= -\oint_{\partial\Omega}\mathcal{P}(r)dS(r) = -\sum_{q\in\mathbb{N}(p)}\mathcal{P}_{pq}\Delta S_{pq}\end{aligned} \tag{48}$$

where $\mathcal{P}_{pq}$ is the average pressure measured on the Voronoi face (ΔS_{pq}) between p and q (see Fig. 1). Considering that $\mathcal{P}_{pq} = \mathcal{P}_{qp}$ but $\Delta S_{pq} = -\Delta S_{qp}$, we simply confirm the anti-commutative nature of the face force as

$$\boldsymbol{F}_{pq} = -\boldsymbol{F}_{qp}, \tag{49}$$

which means that the momentum is conserved by the symmetric nature of the scheme.

4.2. Energy Conservation

The total energy of the Laguerre cell is decomposed into the kinetic and internal energies and the time derivatives of them are related as

$$\frac{dE_p}{dt} = \frac{dT_p}{dt} + \frac{dU_p}{dt} \tag{50}$$

where T and U are the kinetic and internal energies. The change of the kinetic energy is obtained by multiplying equation (48) with the velocity of p as

$$\frac{dT_p}{dt} \equiv \boldsymbol{F}_p \cdot u_p = m_p \frac{du_p}{dt} \cdot u_p \tag{51}$$

$$= -\left(\oint_{\partial\Omega} \mathcal{P}(r) dS(r) \right) \cdot u = \sum_{q \in \mathbb{N}(p)} \mathcal{P}_{pq} \Delta S_{pq} \cdot u_p \tag{52}$$

where the last equation is the discretized sum over the polygon faces. Likewise, the internal energy changes as

$$\frac{dU_p}{dt} = -\oint_{\partial\Omega} \mathcal{P}(r) dS_{pq}(r) \cdot (u^{\rm rad}_{pq}(r) - u_p) \tag{53}$$

$$= -\sum_{q \in \mathbb{N}(p)} \mathcal{P}_{pq} \Delta S_{pq} \cdot (u^{\rm rad}_{pq} - u_p), \tag{54}$$

where $u^{\rm rad}_{pq}$ is the velocity of the Laguerre face, ΔS_{pq}. Then, the volume increase over the face is $\Delta V = S_{pq} \cdot (u^{\rm rad}_{pq} - u_p)$.

Finally, the total energy would be expressed as

$$\frac{dE_p}{dt} = -\sum_{q \in \mathbb{N}(p)} \mathcal{P}_{pq} \Delta S_{pq} \cdot u^{\rm rad}_{pq}. \tag{55}$$

In the calculations this face velocity is $u^{\rm face}_{pq}$ of equation (46), which equals $u^{\rm geo}_{pq}$ on an ordinary face and carries the HLL normal speed only on an extreme face. The change of total energy between p and q is anti-symmetric as

$$\frac{d\boldsymbol{E}_{pq}}{dt} \equiv -\mathcal{P}_{pq} \Delta S_{pq} \cdot u^{\rm rad}_{pq} = -\frac{d\boldsymbol{E}_{qp}}{dt} \tag{56}$$

because $u_{pq} = u_{qp}$ and $S_{pq} = -S_{qp}$. This proves the conservation of total energy of the pair of Voronoi particles consequently.

5. Tests of the geometric-face scheme

GFS is the only scheme used in the tests of this section. Every weight is zero. An ordinary face takes the HLLC pressure and the geometric velocity of equation (41). An extreme face takes the HLL state of equation (44). The integrator is fourth-order Runge–Kutta. An earlier comparison of five face algorithms on the same one-dimensional problems is given in Appendix Appendix F.

5.1. Sod shock tube

The Sod shock tube is the Riemann problem $(\rho, P, u) = (1, 1, 0)$ for $x < 0.5$ and $(0.125, 0.1, 0)$ for $x \geq 0.5$, on $0 \leq x \leq 1$, with reflecting walls. Figure 2 shows GFS at $N = 200$. The density L_1 is 5.86×10^{-3} and the energy error is 1.3×10^{-8}. The shock, the contact, and the rarefaction sit on the reference solution.

5.2. One-dimensional problems

Figure 3 collects eight standard problems at $N = 200$, except the Shu–Osher panel, which uses $N = 800$. Every run is GFS. Transmissive boundaries insert the exterior state when the velocity points into the domain. The density L_1 values are 5.86×10^{-3} (Sod), 0.154 (blast), 1.36×10^{-2} (Noh), 0.571 (Shu–Osher), 5.25×10^{-3} (Lax), 2.86×10^{-2} (double rarefaction), 0.105 (collision), and 6.48×10^{-3} (contact). The energy error is below 10^{-5} on every problem except the blast, where it is 3.3×10^{-2}. In the Shu–Osher problem the left state has velocity 2.629. The shock in the $N = 800$ calculation lies at $x = 2.39$, on the reference shock.

5.3. Gresho vortex

The Gresho vortex is a stationary azimuthal flow on the unit square, with $\rho = 1$ and a triangular $v_\phi(r)$ that peaks at 1 when $r = 0.2$. Figure 4 shows GFS at $t = 1$ with 128×128 particles. The mean absolute error in v_ϕ is 2.1×10^{-2}, and the energy error is 2×10^{-8}.

5.4. KH Instability

The Kelvin–Helmholtz calculation uses the shear layer of Lecoanet et al. on $0 \leq x \leq 1$ and $0 \leq y \leq 2$. The mesh contains 128×256 particles. Every weight is zero, and the face rule is that of Section 3.4. Figure 5 shows the rolled vortices at $t = 2$ and the mixing layer at later times. The integration was stopped at $t = 7.85$. The energy error is -1.9×10^{-7}, and no cell volume is negative.

5.5. Rayleigh–Taylor instability

The Rayleigh–Taylor calculation uses 128×256 particles on $0 \leq x \leq 0.5$ and $0 \leq y \leq 1$. The density ratio is 2 and $g_y = -0.5$. The y boundary is periodic. The integration stops when the dense spike reaches $y = 0$, before the spike re-enters at $y = 1$. Figure 6 shows three spikes. At $t = 4.40$ the minimum y of gas with $\rho > 1.5$ is 0.035. The energy error is $+1.9\times10^{-4}$, and no cell volume is negative. A comparison calculation that assigned the Riemann contact speed to every face, including the shear faces along the spikes, increased the energy by a factor of 10^7 by $t \simeq 2.2$.

5.6. Mach-3 flow past a cylinder

The Mach 3 stream has density $\rho = 1$, velocity $u = 1$, and pressure $P = 0.08$. The sound speed is $\simeq 0.33$. The stream passes a solid cylinder of radius $R = 0.5$ centred at (2.5, 2). Figure 7 uses the domain $0 \leq x \leq 20$ and $0 \leq y \leq 4$ with 400×80 particles, so the cell size is 0.05. The wall is the

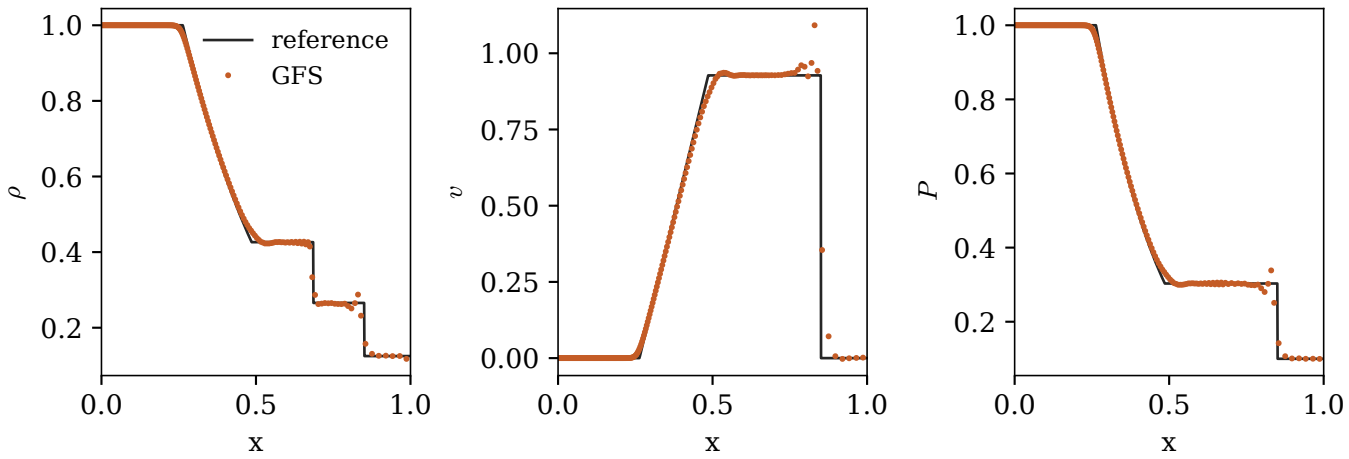

Figure 2. Sod shock tube at the standard end time for GFS with $N = 200$. The panels are density, velocity, and pressure. Points are particles and the line is the reference solution.

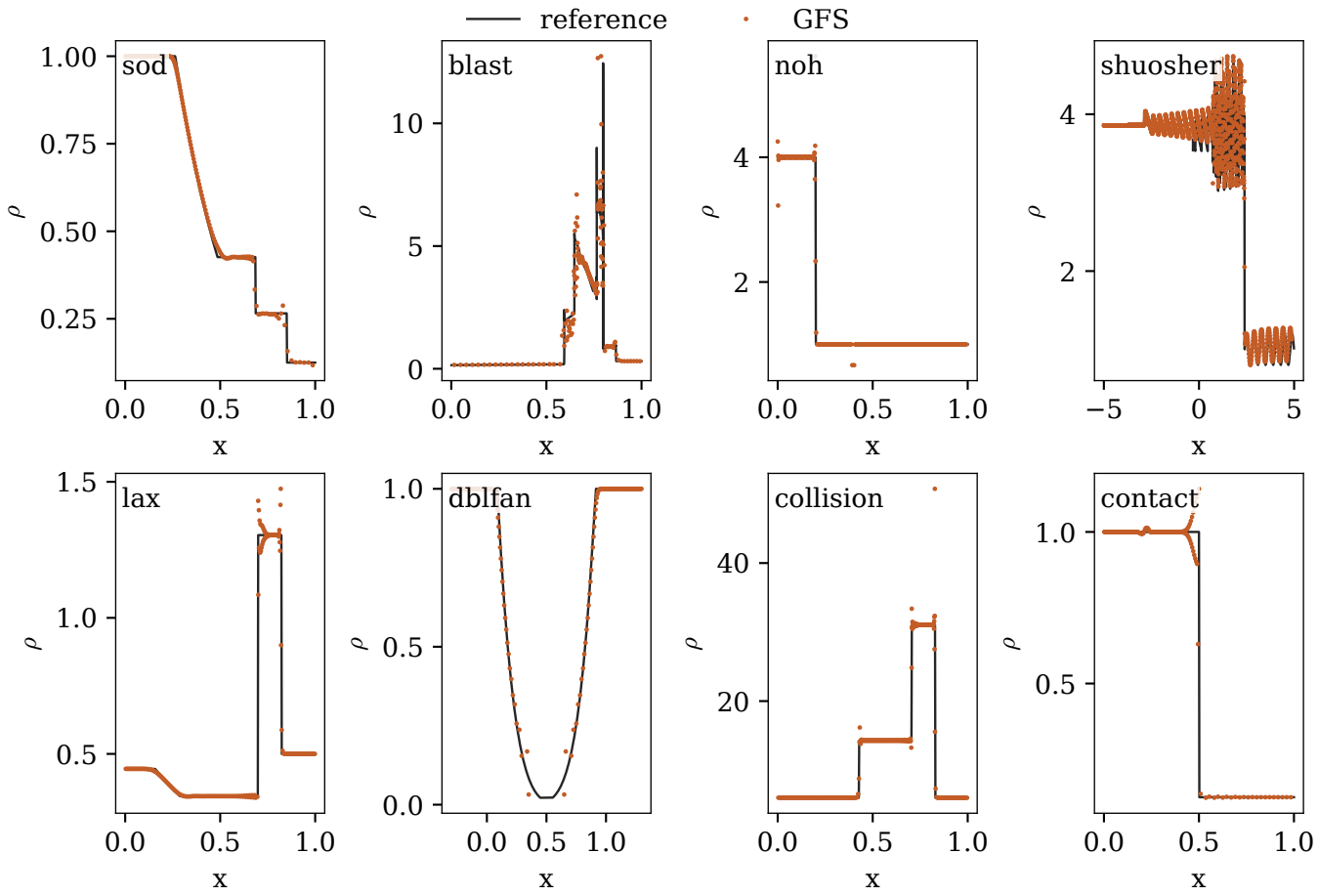

Figure 3. GFS density for eight one-dimensional problems. Points are the particles and the line is the reference. $N = 200$, except Shu–Osher, which uses $N = 800$. The panel title names the problem.

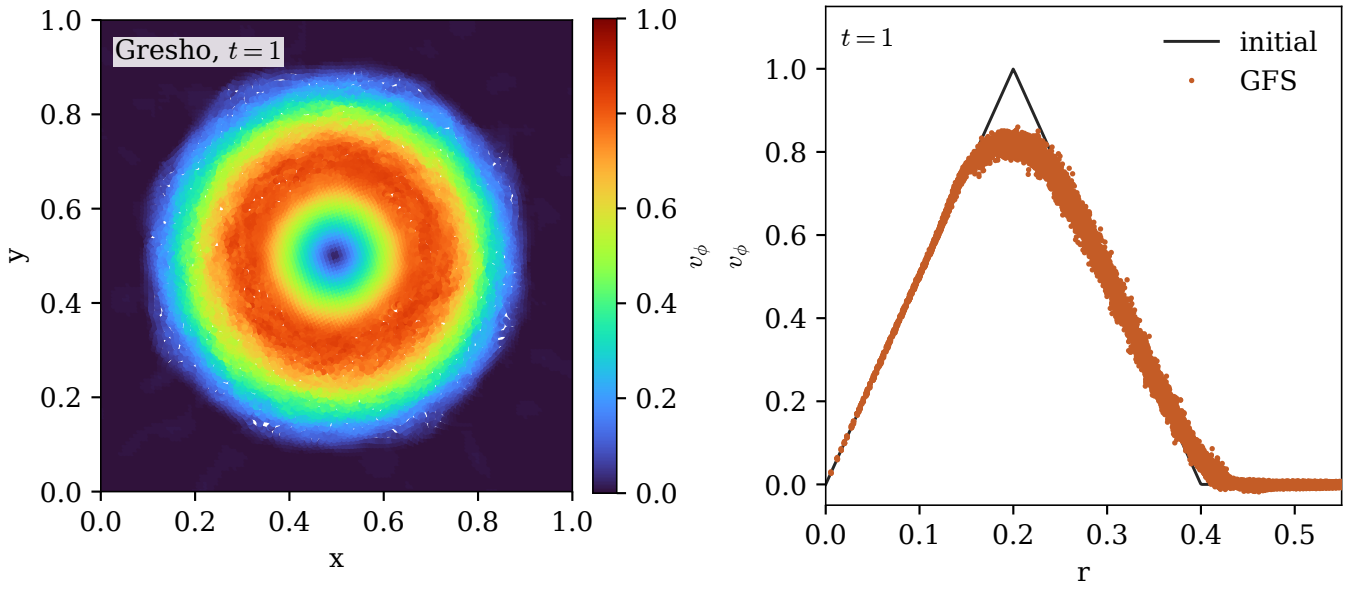

Figure 4. Gresho vortex with GFS at $t = 1$. The left panel is v_ϕ. The right panel compares the particles with the initial profile. The box is $0 \leq x, y \leq 1$.

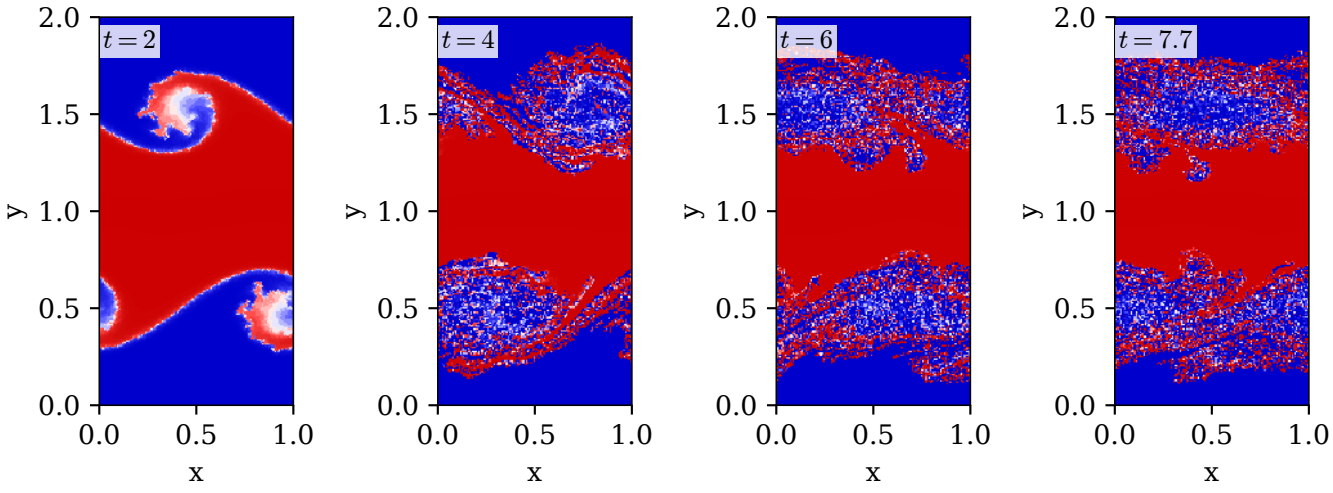

Figure 5. Kelvin–Helmholtz density on $0 \leq x \leq 1$, $0 \leq y \leq 2$, with 128×256 particles and GFS. The primary rolls are visible at $t = 2$. By $t = 7.7$ they have broken into a mixing layer while the dense core of the stripe remains.

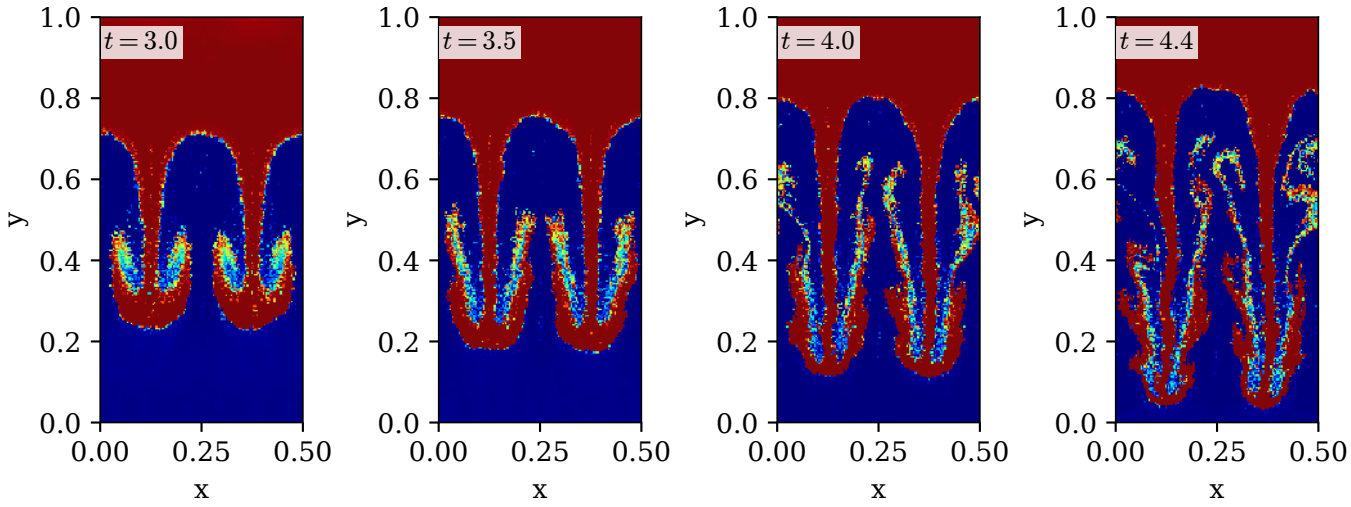

Figure 6. Rayleigh–Taylor density with GFS. Heavy fluid is on top. The three spikes lengthen from $t = 3.0$ ($y_{\rm min} = 0.228$ for $\rho > 1.5$) to $t = 4.4$ ($y_{\rm min} = 0.035$).

reflecting ghost in equation (47). At $t = 5$ no particle lies inside the cylinder, and the smallest particle radius is 0.504. The energy differs from its initial value by -1.0%. The maximum density is 9.4, and the time step is $\simeq 8\times10^{-3}$. A previous integration that projected particles onto $r = R$ after each step reached $\rho_{\rm max} = 635$ and an energy error of order 10^7 by $t = 2.58$. The black disk in the figure is the solid cylinder. The thin white edge is a plotting stroke that keeps the disk visible against the wake. The stroke is not a layer of absent gas.

6. Astrophysical Recipes

6.1. Gas Cooling and Heating

6.2. IMF of Stars

6.3. Stellar and SN feedbacks

6.4. Metallicity Enrichment

6.5. AGN Feedbacks

7. Discussion

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Appendix A Weight tracers not used by GFS

A.1 How to Evolve the Power Distance

To actively adjust in the site's power distance (ω), we apply a model of the expansion or contraction of a gas cell to scale with the enthalpy as $\omega^2 \propto \mathcal{H}_i = mC_s^2/(\gamma - 1)$. Therefore, the change of power distance is

$$\Delta\omega^i \equiv \tilde{\omega}^{i+1} - \omega^i = \left[\left(\frac{\mathcal{H}^i}{\langle\mathcal{H}^i\rangle}\right)^{1/2} - 1\right]\omega^i, \qquad \text{(A1)}$$

where $\langle \mathcal{P}^i \rangle$ is defined as surface-averaged pressure of surrounding neighbors. It is noteworthy that during the IC setup,

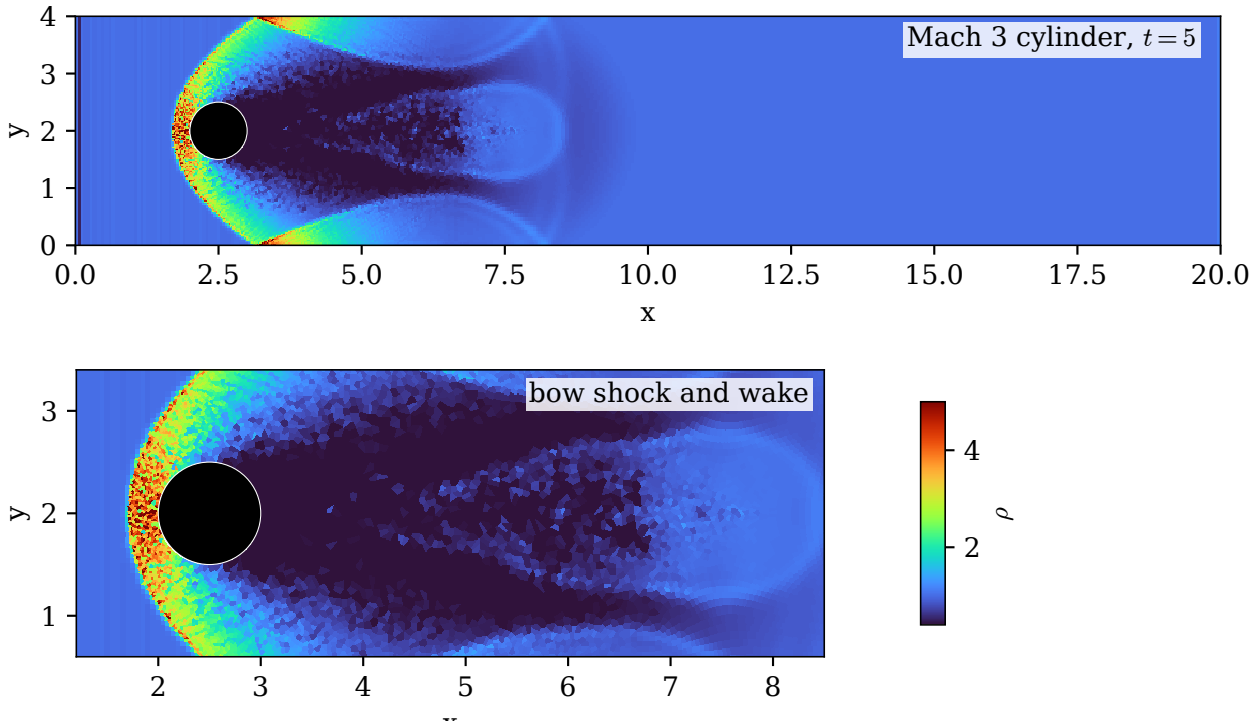

Figure 7. Mach-3 flow past a solid cylinder at $t = 5$ on $0 \le x \le 20$. The black disk is the cylinder. The bow shock stands upstream and the wake extends downstream. The colour scale runs from 0.4 to 5.

a separate scaling parameter, $\langle\omega\rangle = \varkappa_\omega d_{\rm mean}$, is introduced, typically initialized with $\varkappa_\omega = 0.5$.

However, since the rate of power distance change ($v_p^\omega \equiv \Delta\omega_p/\Delta t$) should not exceed the sound speed, the maximum expansion or contraction of the power distance per time step is constrained by

$$-c_s(p)\Delta t \le \Delta\omega_p \le c_s(p)\Delta t \tag{A2}$$

where $c_s(p)$ is the sound speed and Δt is the time step size. The value of power radius at step i is, consequently, the minimum among the two corresponding values as

$$\omega_p^i = \min\left(\tilde{\omega}_p^i, \omega_{p,\rm ceil}^i\right). \tag{A3}$$

A.2 Thermodynamic Tracers for the Power Distance

The enthalpy-based model above treats the power distance as a response to the local thermodynamic state. Because the Laguerre tessellation is invariant under a rigid rescaling $\omega_p \to \omega_p + c$, what physically controls the geometry is the *variation* of ω^2 across the flow, not its absolute value. This invites a family of alternatives differing only in the tracer functional Φ that is used to weight the sites. We implement three such options, all of the form

$$\omega_p^2 = \varkappa_\omega^2\, d_{\rm mean}^2\, q_p, \qquad q_p = Q\big(\mathcal{P}_p, \rho_p\big), \tag{A4}$$

with $\varkappa_\omega\, d_{\rm mean}$ setting the overall scale and Q a dimensionless profile. The implemented tracers are:

(i) Pressure-based (default, `w2_mode=0`). $Q = \left(\mathcal{P}_p/\mathcal{P}_0\right)^{\alpha_\omega}$ with $\alpha_\omega = (\gamma-1)/\gamma$ and $\mathcal{P}_0$ a reference pressure fixed by the IC. At a subsonic contact discontinuity ($\mathcal{P}_p = \mathcal{P}_q$) this gives $\omega_p^2 \approx \omega_q^2$, so the radical axis coincides with the Voronoi bisector and cells across the contact share a common size. This is the choice that dissipates least at material interfaces but carries no material tag.

(ii) Entropy-A power law (`w2_mode=1`). $Q = \left(A_p/A_{\rm ref}\right)^{\alpha_\omega}$ with the entropic function $A_p \equiv \mathcal{P}_p/\rho_p^\gamma$ and $A_{\rm ref} = \mathcal{P}_0/\rho_{\rm ref}^\gamma$ (geometric-mean density in the IC). In smooth adiabatic flow A_p is conserved along streamlines, so ω_p is constant within each fluid but jumps across contacts by $(\rho_q/\rho_p)^{\alpha_\omega\gamma}$. Cells of different materials therefore have geometrically distinct sizes—a useful tag, but the contact acquires a face-area asymmetry that can bias the flux calculation when α_ω is large.

(iii) Specific-entropy additive (`w2_mode=2`). $Q = 1 + \beta\,(s_p - s_{\rm ref})/s_{\rm scale}$, where $s_p \equiv c_v \ln(\mathcal{P}_p/\rho_p^\gamma)$ is the specific entropy (up to the constant c_v, which cancels through $s_{\rm scale}$), $s_{\rm scale} = |s_1 - s_2| = \gamma|\ln(\rho_1/\rho_2)|$ is a scale set by the IC contact jump, and $\beta \in [0, 1)$ is a dimensionless amplitude. Unlike (ii), this tracer is *linear* in s and bounded: the perturbation to ω^2 is bounded by β and so is the contact jump in cell size. The advective stability of s in smooth flow means the noise feedback $\delta\rho \to \delta A \to \delta\omega \to \delta\rho$ is damped both

logarithmically (through $s = c_v \ln A$) and linearly (through β), making small-β runs safe against the feedback-driven cell shrinkage that can plague (ii). With the Kelvin–Helmholtz IC ($\rho_1 = 1$, $\rho_2 = 2$, $\gamma = 5/3$), for example, $s_{\rm scale} = \gamma \ln 2 \approx 1.155$ and $\beta = 0.15$ produces an $\approx 11\%$ contrast in ω^2 across the contact, strong enough to retain material identity while remaining within the linear-perturbation regime.

For all three tracers the time-step safeguard of equation (36) applies unchanged, and the ceiling $\omega_{p,\rm ceil}$ together with the optional relaxation, rate, and floor limiters (τ_ω, r_ω, $\omega_{\rm floor}$) act identically on the resulting q_p. The choice of tracer therefore controls only the functional form, not the admissible range, of the Laguerre weight.

Appendix B Face algorithms not used by GFS

The algorithms in this appendix were implemented and compared during the development of the code. They are not part of GFS.

Appendix C Face-averaged Interpolations

C.1 A Relation between Cell-Averaged and Face-Averaged values

First, we use the well-designed interpolation to get the average surface value from the corresponding Voronoi sites as

$$\langle \phi \rangle_{\rm face} \equiv \frac{1}{\partial\Omega(\mathbb{N}_n)} \int_{\partial\Omega(\mathbb{N}_n)} \phi dS = \sum_{i\in\mathbb{N}_n} w_i \langle \phi_i \rangle_{\rm cell} \tag{C1}$$

where $\mathbb{N}_n$ is the Voronoi sites shaping the face of interest and w_i is the corresponding weighting. Then, according to the Gauss theorem, we may get the spatial difference of ϕ from the surface integral as,

$$\langle \nabla\phi \rangle_{\rm cell} = \frac{1}{\Omega_{\rm cell}} \oint_{\partial\Omega} \phi dS = \frac{1}{\Omega_{\rm cell}} \sum_{\rm face} \langle \phi \rangle_{\rm face} \Delta S_{\rm face} \tag{C2}$$

where $\langle \phi \rangle_{\rm face}$ is the averaged value on the face element, $\Delta S_{\rm face}$, and $\Omega_{\rm cell}$ is the cell volume.

To summarize, equation (C3) shows how the derivatives of the cell-averaged value are obtained by applying the recursive loop.

$$\begin{aligned} \langle \phi \rangle_{\rm cell} \to \langle \phi \rangle_{\rm face} \Rightarrow \langle \nabla\phi \rangle_{\rm cell} \to \\ \to \langle \nabla\phi \rangle_{\rm face} \Rightarrow \left\langle \nabla^2\phi \right\rangle_{\rm cell} \to \\ \to \left\langle \nabla^2\phi \right\rangle_{\rm face} \Rightarrow \left\langle \nabla^3\phi \right\rangle_{\rm cell} \to \cdots \end{aligned} \tag{C3}$$

where $\Rightarrow$ means the cell-to-face interpolation and $\to$ means applying of the surface integral (eq. C2). In the next subsequent sections, we will show how to compute those values in detail.

C.2 Effects of the Simple Averaging Scheme

A hydro quantity (ϕ) on the cell center is used to get the face-centered value on a Laguerre surface. For example, the second-order accuracy between p and q is simply given as,

$$\phi_{pq} = \frac{1}{2}\left(\phi_p + \phi_q\right), \tag{C4}$$

which satisfies the symmetry of $\phi_{pq} = \phi_{qp}$.

If we simply adopt the average velocity on the Laguerre face as $u_{pq} = \frac{1}{2}(u_p + u_q)$, we may get a simple relation as

$$\frac{dU_p}{dt} = -\frac{1}{2} \sum_{q\in\mathbb{N}(p)} \mathcal{P}_{pq} \Delta S_{pq} \cdot (u_q - u_p) \tag{C5}$$

$$= -\frac{1}{2} \sum_{q\in\mathbb{N}(p)} \Delta U_{pq}, \tag{C6}$$

which is symmetric between p and q. Therefore, it implies that the change of internal energy is same between the Laguerre pair ($\Delta U_{pq} = \Delta U_{qp}$). Also another intriguing property can be obtained when the pressure on the Laguerre faces are averaged as $P_{pq} = (P_p + P_q)/2$ and, then, we may obtain from equation (48) as

$$F_p = -\sum_{q\in\mathbb{N}(p)} \left(\frac{\mathcal{P}_q}{2}\right) \Delta S_{pq} \tag{C7}$$

where we have used the relation $\sum_q \Delta S_{pq} = 0$. From this result, we may know that the force on the target particle is completely determined by the pressure of its Laguerre neighbors.

C.3 Generalization of the Face-Centered Approximation

For the higher-order generalization, we adopt the functional form of equation (C4) as

$$\phi_{pq} = \sum_{s\in\mathbb{N}(p)} W^s_{pq} \phi_s \tag{C8}$$

where ϕ_s is the cell-centered hydro quantity (pressure or velocity) of the neighbor, s, which may have a contribution to forming the Voronoi surface, ΔS_{pq}.

The weighting for the neighboring particles is expressed into m-dimensional vector as

$$\boldsymbol{W}_{pq} \equiv (w_p, w_q, w_r \cdots, w_{r+m-3}) \tag{C9}$$

where the first element is the contribution from the target site, the second one is from the facing neighbor, and the remains are contributions from the other neighbors. For the conservation of momentum and energy, we have to set $w_p = w_q$ and $w_r = \cdots = w_{r+m-3}$.

C.4 Weighting on One-Dimensional Surface

In the 2-dimensional space, the weighting vector should have four components because to make a Laguerre surface we need three neighboring sites beside the target site (see

Fig. C1). Below, we show two exemplary cases for the two-dimensional weighting vector;

$$\boldsymbol{W}^{\rm 2D}_{\rm 2nd} = \left(\frac{1}{2}, \frac{1}{2}, 0, 0\right) \tag{C10}$$

$$\boldsymbol{W}^{\rm 2D}_{\rm LP} = \left(\frac{4}{9}, \frac{4}{9}, \frac{1}{18}, \frac{1}{18}\right), \tag{C11}$$

where the first one has the second-order accuracy while the last one has the fourth-order accuracy (Lilek & Perić 1995). Figure C1 depicts how to formulate the weighting of the contributions of neighbors in 2-D. As the central node on the 2-D Laguerre face is equidistant from p and q, we may set $\phi_c = (\phi_p + \phi_q)/2$. Likewise, for the left- and right-end nodes of the Laguerre surface, we set $\phi_1 = (\phi_p + \phi_q + \phi_s)/3$ and $\phi_2 = (\phi_p + \phi_q + \phi_r)/3$, respectively because the end node is equidistant from those three sites. We can get the average value on the Laguerre face as

$$\overline{\phi} = (1 - 2\omega)\,\phi_c + \omega\phi_{\rm l} + \omega\phi_{\rm r} \tag{C12}$$

$$= \left(\frac{1}{2} - \frac{\omega}{3}\right)(\phi_p + \phi_q) + \frac{\omega}{3}(\phi_r + \phi_s) \tag{C13}$$

where ω is the weighting of the edge node, and it should have a range as $0 \le \omega \le 1/2$. If the edge nodes have a null contribution to the average value on the 2-D Laguerre face, then we may set $\omega = 0$ and this is the case for $\boldsymbol{W}^{\rm 2D}_{\rm 2nd}$. For $\boldsymbol{W}^{2D}_{\rm LP}$, we get $\omega = 1/6$. See Appendix I for the affine combination of nodes with the homogeneous interpolation on the line.

C.5 Weighting on Two-Dimensional Surface

On the two-dimensional surface, the situation becomes more complex, but it can be managed by decomposing the Laguerre face into multiple triangles. Since the Laguerre face forms a convex polygon in 3-D, we may find the barycenter of the polygon and interpolate values within each triangle.

`https://www.inf.usi.ch/hormann/papers/Hormann.2014.BI.pdf`

`Mean Value Coordinates for Arbitrary Planar Polygons KAI HORMANN Clausthal University of Technology`

To illustrate the adopted weighting process, we present Figure C2. First, in panel (a), the Laguerre face is highlighted in gray. This face is a pentagon, with its barycenter indicated by the symbol O in panel (c). The pentagon is tessellated into five triangles, all sharing the barycenter as a common vertex. A shear mapping is applied to each triangle to transform them to right triangles, as shown in the transition from panel (c) to panel (d).

According to the model of two-dimensional univariate barycentric interpolation, we define values at arbitrary positions inside the triangle as

$$\phi(x, y) \equiv \frac{\alpha\phi_A + \beta\phi_B + \gamma\phi_O}{\alpha + \beta + \gamma} \tag{C14}$$

where α, β, and γ are the surface areas of subtriangles corresponding to their vertices (see panel (c)). The normalized

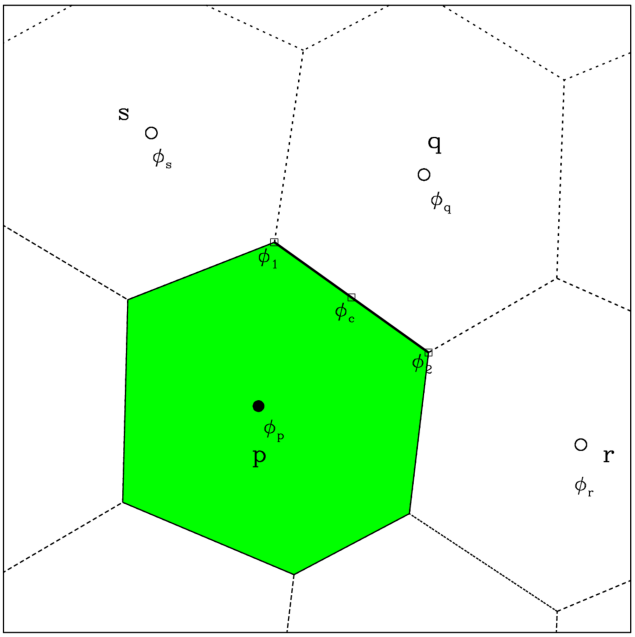

Figure C1. Illustration of two-dimensional weighting scheme. The target site (p) marked with the filled circle has a cell color-coded with green. Three neighboring sites (q, r, and s) marked with open circles forms the Laguerre face (thick line). Here we mark the central and left-/right-end nodes of the Laguerre face with the open squares. The values of ϕ on the face are denoted as ϕ_1, ϕ_c, and ϕ_2, from the left.

terms of them are often called the barycentric coordinate for (x, y). Then, for example, the contribution of the vertex A to the point (x, y) is proportional to the area of subtriangle, α.

On the other hand, after the shear mapping as show in panel (d), we are able to measure the average value on the triangle as

$$\overline{\phi}_{\Delta ABO} = \int_{\Delta ABO} P(x, y)\,dx\,dy \tag{C15}$$

$$= \frac{1}{3}\left(\phi_O + \phi_A + \phi_B\right). \tag{C16}$$

which simply implies that the average value on the triangle is a simple average of values defined at the vertices. While the value at the barycenter is determined by the two sites as $\phi_O = (\phi_p + \phi_q)/2$, the values at the other two nodes are averaged by additional neighbors as

$$\phi_A = \frac{1}{4}\left(\phi_p + \phi_q + \phi_r + \phi_s\right) \tag{C17}$$

$$\phi_B = \frac{1}{4}\left(\phi_p + \phi_q + \phi_t + \phi_u\right). \tag{C18}$$

From now on, rather than using equation (C16), we adapt the relative weighting of nodes as

$$\overline{\phi}_{\Delta ABO} = (1 - 2\omega)\,\phi_O + \omega\phi_A + \omega\phi_B \tag{C19}$$

$$= \left(\frac{1}{2} - \frac{\omega}{2}\right)(\phi_p + \phi_q) + \frac{\omega}{4}\left(\phi_r + \cdots + \phi_u\right)$$

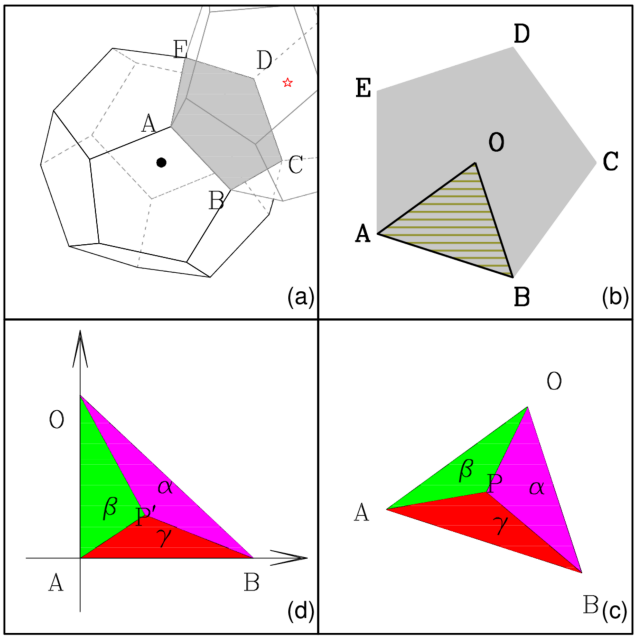

Figure C2. Illustration of the three-dimensional weighting scheme. Our approach for determining the average value on the Laguerre surface (represented as a gray pentagon) is depicted in a counterclockwise sequence, beginning from the top-left panel. (a) Two Laguerre cells are shown, with the shared Laguerre face highlighted in gray. (b) The Laguerre surface is decomposed into multiple triangles, and one example triangle is shown. (c) With the arbitrary inner point P, the triangle is further tessellated with three subtriangles with areas, α, β, and γ, which are called the barycenter coordinate of P after normalization. (d): Through a proper shear mapping, the triangle can be transformed to a right triangle. In this mapping, the area is preserved and straight lines remain straight.

For $\omega = 1/3$, we may get the weighting vector on the triangle as

$$\boldsymbol{W}_{\rm JK}^{\rm 3D} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}\right) \tag{C20}$$

Finally, the surface-weighted average of ϕ on the tessellated triangles is obtained as

$$\langle\phi\rangle_{\rm VF} = \frac{\sum_i S(\Delta_i)\overline{\phi}_i}{\sum_i S(\Delta_i)}$$

where $S(\Delta_i)$ is the area of the i'th tessellated triangle and $\overline{\phi}_i$ is the average ϕ on the triangle.

C.6 Face-Averaged Viscous Stress

The Navier–Stokes viscous stress tensor established in Section 2 (eq. 20) can be naturally incorporated into the M(n, m) face-pressure framework without additional machinery. The cell-averaged velocity gradient tensor is already available from Section 2.8:

$$\langle\boldsymbol{\nabla u}\rangle_p = \frac{1}{\Omega_p}\sum_{q\in\mathbb{N}(p)} u_{pq}\otimes\Delta S_{pq}, \tag{C21}$$

together with the cell-averaged divergence

$$\langle\nabla\cdot u\rangle_p = \frac{1}{\Omega_p}\sum_{q\in\mathbb{N}(p)} u_{pq}\cdot\Delta S_{pq}, \tag{C22}$$

where u_{pq} is the face-averaged velocity from the M(n, m) interpolation of Section 5.3.

The cell-averaged viscous stress tensor follows directly:

$$\langle\boldsymbol{\tau}\rangle_p = -\nu_p\rho_p\left(\langle\boldsymbol{\nabla u}\rangle_p + \langle\boldsymbol{\nabla u}\rangle_p^{\mathsf{T}} - \tfrac{2}{3}\boldsymbol{I}\,\langle\nabla\cdot u\rangle_p\right). \tag{C23}$$

The face-averaged viscous stress is obtained via the same M(n, m) weighting used for pressure and velocity:

$$\langle\boldsymbol{\tau}\rangle_{pq} = \sum_s W_{pq}^s\,\langle\boldsymbol{\tau}\rangle_s\,, \tag{C24}$$

inheriting the symmetry $\langle\boldsymbol{\tau}\rangle_{pq} = \langle\boldsymbol{\tau}\rangle_{qp}$ from the M(n, m) weights.

The momentum equation (48) generalizes to

$$m_p\frac{du_p}{dt} = -\sum_{q\in\mathbb{N}(p)}\left(\langle\mathcal{P}\rangle_{pq}\,\boldsymbol{I} + \langle\boldsymbol{\tau}\rangle_{pq}\right)\cdot\Delta S_{pq}, \tag{C25}$$

and the scalar viscous pressure (normal component) relevant for the face force is

$$\Pi_{pq}^{\tau} = \hat{n}_{pq}\cdot\langle\boldsymbol{\tau}\rangle_{pq}\cdot\hat{n}_{pq}. \tag{C26}$$

Physical and artificial viscosity.—The kinematic viscosity ν_p in equation (C23) admits two interpretations. For physical (molecular) viscosity, ν is a constant determined by the Reynolds number of the flow. For shock capturing, ν_p is promoted to a cell-dependent artificial viscosity:

$$\nu_p = \alpha_p\,h_p\,c_p, \tag{C27}$$

where $h_p = (\Omega_p)^{1/d}$ is the effective cell size, c_p is the sound speed, and α_p is the viscosity coefficient. With constant α this recovers the classical Monaghan-type artificial viscosity; with the time-dependent CD10 switch (Section 6.5, eqs. M10–M11), the stress tensor adapts automatically to shocks.

Connection to signal-velocity AV.—In one dimension with uniform spacing $\Delta x \approx h$, the NS stress tensor reduces to $\tau \approx -\alpha hc\,\rho\,(2\,\partial u/\partial x)$. Comparing with the Price signal-velocity AV, $\Pi = \frac{1}{2}\alpha\,v_{\rm sig}\,\bar{\rho}\,\Delta u$, and noting $v_{\rm sig} \approx 2c$ for weak waves, both expressions scale as $\alpha c\rho\Delta u$. The NS form is the natural tensorial generalization, correctly distinguishing shear from compression in multiple dimensions.

C.7 Time Steps

The hydrodynamic time step is determined from the sound speed and the radial velocity as

$$\Delta t_{pq} = 2C_{\rm cfl}\left(\frac{\|\boldsymbol{r}_{pq}\|}{v_{\rm sig}}\right) \tag{C28}$$

where $v_{\rm sig} \equiv \left(c_p^s + c_q^s - \min(0, u_{pq}\cdot e_{pq})\right)$ and $C_{\rm cfl}$ is the Courant number. Finally, the global time step is determined as

$$\Delta t = \min_p\left(\min_{q\in\mathbb{N}(p)}(\Delta t_{pq})\right) \tag{C29}$$

For the fourth-order Runge-Kupta integration, we set $c_{\rm cfl} = 0.2$ while we use $c_{\rm cfl} = 0.1$ for the Leap-frog scheme.

Appendix D The HLLC + Cullen–Dehnen Scheme on 3D Laguerre Cells

D.1 Motivation

A simple artificial viscosity employs a fixed pair of parameters (α, β) applied uniformly to every face. While adequate for moderate shocks, this global prescription overdissipates smooth flows and under-resolves contact discontinuities. We therefore introduce a Riemann-solver–based face pressure combined with a time-dependent viscosity switch, both validated in one dimension (Appendix M). This section describes how the algorithm maps onto the cell-averaged / face-averaged framework established in Sections 4 and 5.

In this framework every hydrodynamic quantity is either a cell-averaged value $\langle\phi\rangle_{\rm cell}$, volume-integrated over the Lagrangian cell Ω_p, or a face-averaged value $\langle\phi\rangle_{\rm face}$, area-integrated over the Laguerre face $\partial\Omega_{pq}$. The recursive loop of equation (C3) connects the two:

$$\langle\phi\rangle_{\rm cell} \xrightarrow{\text{interpolation}} \langle\phi\rangle_{\rm face} \Longrightarrow{\text{surface integral}} \langle\nabla\phi\rangle_{\rm cell}\,.$$

The HLLC+CD10 scheme replaces the interpolation step ($\rightarrow$) for the pressure and normal velocity with a Riemann-solver–based procedure, while the surface-integral step ($\Rightarrow$) and the M(n, m) weighting for tangential quantities remain unchanged.

D.2 Cell-Averaged Gradient via the Green–Gauss Theorem

The starting point is the cell-averaged gradient already established in equation (C2):

$$\langle\nabla\phi\rangle_p = \frac{1}{\Omega_p}\sum_{q\in\mathbb{N}(p)}\langle\phi\rangle_{pq}\,\Delta S_{pq}, \tag{D1}$$

where $\langle\phi\rangle_{pq}$ is the face-averaged value on $\partial\Omega_{pq}$. For the initial gradient estimate we use the second-order face average of equation (C4), $\langle\phi\rangle_{pq} = (\langle\phi\rangle_p + \langle\phi\rangle_q)/2$, so that equation (D1) becomes

$$\langle\nabla\phi\rangle_p = \frac{1}{2\Omega_p}\sum_{q\in\mathbb{N}(p)}\left(\langle\phi\rangle_q - \langle\phi\rangle_p\right)\Delta S_{pq}, \tag{D2}$$

where we used the closure relation $\sum_q \Delta S_{pq} = 0$. This is identical to the cell-averaged velocity gradient of Section 2.8 and constitutes the first step of the recursive loop (C3).

Equation (D2) is applied to $\phi \in \{P, \rho, u\}$ to obtain the cell-averaged gradients $\langle\nabla P\rangle_p$, $\langle\nabla\rho\rangle_p$, and $\langle\nabla u\rangle_p$.

D.3 Slope Limiter for Cell-Averaged Gradients

The unlimited gradient (D2) can create new extrema when extrapolated to face centroids. To enforce monotonicity we apply the Barth–Jespersen limiter (Barth & Jespersen 1989). For each cell p, define the extrapolated increment toward face pq:

$$\delta\phi_{pq} = \langle\nabla\phi\rangle_p\cdot(\mathbf{x}_{f,pq} - \mathbf{x}_p), \tag{D3}$$

where $\mathbf{x}_{f,pq}$ is the area-weighted centroid of the Laguerre face $\partial\Omega_{pq}$ and $\mathbf{x}_p$ is the cell centroid. The neighborhood extrema are

$$\phi_{\max} = \max_{q\in\mathbb{N}(p)\cup\{p\}}\langle\phi\rangle_q\,, \qquad \phi_{\min} = \min_{q\in\mathbb{N}(p)\cup\{p\}}\langle\phi\rangle_q\,. \tag{D4}$$

The per-face limiter value is

$$\Phi_{pq} = \begin{cases} \min\left(1,\ \dfrac{\phi_{\max} - \langle\phi\rangle_p}{\delta\phi_{pq}}\right), & \delta\phi_{pq} > 0, \\ \min\left(1,\ \dfrac{\phi_{\min} - \langle\phi\rangle_p}{\delta\phi_{pq}}\right), & \delta\phi_{pq} < 0, \\ 1, & \delta\phi_{pq} = 0, \end{cases} \tag{D5}$$

and the cell limiter is $\Phi_p = \min_{q\in\mathbb{N}(p)}\Phi_{pq}$. The limited cell-averaged gradient is then

$$\widetilde{\langle\nabla\phi\rangle}_p = \Phi_p\,\langle\nabla\phi\rangle_p\,. \tag{D6}$$

This replaces the 1-D minmod limiter (Appendix M) with a multi-dimensional analog that operates directly on the cell-averaged gradient vector.

D.4 MUSCL-Extrapolated Face States

Using the limited gradient (D6), we extrapolate the cell-averaged values to each face centroid to obtain left and right face states:

$$\langle\phi\rangle_L = \langle\phi\rangle_p + \widetilde{\langle\nabla\phi\rangle}_p\cdot(\mathbf{x}_{f,pq} - \mathbf{x}_p), \tag{D7}$$

$$\langle\phi\rangle_R = \langle\phi\rangle_q + \widetilde{\langle\nabla\phi\rangle}_q\cdot(\mathbf{x}_{f,pq} - \mathbf{x}_q). \tag{D8}$$

This procedure is applied to the pressure and density, yielding (P_L, ρ_L) and (P_R, ρ_R), and to the velocity projected onto the face normal:

$$u_{L,n} = \langle u\rangle_L\cdot\hat{n}_{pq}, \qquad u_{R,n} = \langle u\rangle_R\cdot\hat{n}_{pq}, \tag{D9}$$

where $\hat{n}_{pq} = e_{pq} = (q-p)/\|q-p\|$ is the unit normal to the Laguerre face. The sound speeds are $c_L = \sqrt{\gamma P_L/\rho_L}$ and likewise for c_R.

In the language of Section 5, equations (D7)–(D8) replace the M(n, m) interpolation for the variables entering the Riemann problem. The M(n, m) weighting ($\boldsymbol{W}_{\rm JK}^{\rm 3D}$ of Section 5.4) is retained for the tangential velocity and for any face-averaged quantities not entering the Riemann solve.

D.5 HLLC Riemann Solver on Laguerre Faces

Given the MUSCL-extrapolated face states $(P_L, u_{L,n}, \rho_L, c_L)$ and $(P_R, u_{R,n}, \rho_R, c_R)$, the HLLC solver of Appendix M is applied identically. The face-averaged star-state pressure and contact velocity are:

$$\langle P^* \rangle_{pq} = \frac{W_R P_L + W_L P_R + W_L W_R (u_{L,n} - u_{R,n})}{W_L + W_R}, \quad \text{(D10)}$$

$$u_n^* = \frac{W_L u_{L,n} + W_R u_{R,n} + P_L - P_R}{W_L + W_R}, \quad \text{(D11)}$$

where $W_K = \rho_K c_K q_K$ with

$$q_K = \begin{cases} \sqrt{1 + \dfrac{\gamma+1}{2\gamma}\left(\dfrac{P_{\rm pvrs}}{P_K} - 1\right)}, & P_{\rm pvrs} > P_K, \\ 1, & \text{otherwise}, \end{cases} \quad \text{(D12)}$$

and $P_{\rm pvrs}$ the linearized pressure estimate (see Appendix M).

D.1 Composing the 3D Face-Averaged Velocity

The HLLC solver provides only the normal component u_n^* of the face velocity. The full face-averaged velocity is composed by replacing the normal component of the Laguerre face velocity (eq. 39) with the HLLC result, while preserving the tangential and rotational parts:

$$\boxed{\langle u \rangle_{pq}^{\rm rad} = \underbrace{u_n^* \, \hat{n}_{pq}}_{\text{HLLC (normal)}} + \underbrace{\langle \bar{u}_\perp \rangle_{pq}}_{\text{tangential}} + \underbrace{\langle u_{\rm rot} \rangle_{pq}}_{\text{rotational}},} \quad \text{(D13)}$$

where the face-averaged tangential velocity is

$$\langle \bar{u}_\perp \rangle_{pq} = \frac{1}{2}\left[(\langle u \rangle_p + \langle u \rangle_q) - \left((\langle u \rangle_p + \langle u \rangle_q) \cdot \hat{n}_{pq}\right)\hat{n}_{pq}\right], \quad \text{(D14)}$$

and the face-averaged rotational velocity from Appendix D is

$$\langle u_{\rm rot} \rangle_{pq} = \frac{\hat{n}_{pq} \times (\langle u \rangle_q - \langle u \rangle_p)}{\|q - p\|} \times (\mathbf{x}_{f,pq} - \mathbf{m}_{pq}), \quad \text{(D15)}$$

with $\mathbf{m}_{pq}$ the midpoint of the radical plane.

This decomposition ensures that (i) the normal-direction wave structure is resolved by the Riemann solver, (ii) the tangential transport retains the symmetry $\langle \bar{u}_\perp \rangle_{pq} = \langle \bar{u}_\perp \rangle_{qp}$, and (iii) angular momentum is conserved through the rotational component.

D.6 Cullen–Dehnen Viscosity Switch in 3D

D.1 Cell-Averaged Divergence

The cell-averaged velocity divergence is obtained from the face-averaged velocity via the surface integral (eq. C2):

$$\langle \nabla \cdot u \rangle_p = \frac{1}{\Omega_p} \sum_{q \in \mathbb{N}(p)} \langle u \rangle_{pq} \cdot \Delta S_{pq}, \quad \text{(D16)}$$

which is the trace of the cell-averaged velocity gradient tensor of Section 2.8. Here, $\langle u \rangle_{pq}$ is the face-averaged velocity from the second-order average (eq. C4).

D.2 Effective Cell Size

The effective cell size for the CD10 switch is the sphere-equivalent radius:

$$h_p = \left(\frac{3\,\Omega_p}{4\pi}\right)^{1/3}. \quad \text{(D17)}$$

D.3 Alpha Evolution

The per-cell viscosity parameter α_p evolves following the same rule as the 1-D case (Appendix Appendix M, eqs. M10–M11). The compression indicator uses the cell-averaged divergence:

$$A_p = \max\left(0, \; -\frac{\langle \nabla \cdot u \rangle_p^{(n)} - \langle \nabla \cdot u \rangle_p^{(n-1)}}{\Delta t}\right), \quad \text{(D18)}$$

and the target viscosity coefficient is

$$\alpha_{\rm loc} = \alpha_{\max} \frac{h_p^2 A_p}{v_{s,p}^2 + h_p^2 A_p}, \quad \text{(D19)}$$

with asymmetric update: instant rise on compression, exponential decay with timescale $\tau_p = h_p/(2\ell\, v_{s,p})$ otherwise. The maximum signal velocity is accumulated over all faces of cell p:

$$v_{s,p} = \max_{q \in \mathbb{N}(p)} \left(c_p + c_q + \beta \, \max(\Delta u_n, 0)\right), \quad \text{(D20)}$$

where $\Delta u_n = (\langle u \rangle_p - \langle u \rangle_q) \cdot \hat{n}_{pq}$.

D.4 Face-Averaged Viscosity Pressure

The viscous contribution to the face-averaged pressure is applied only in compression ($\Delta u_n > 0$):

$$\langle \Pi \rangle_{pq} = \frac{1}{2} \, \bar{\alpha}_{pq} \, v_{\rm sig} \, \bar{\rho}_{pq} \, \Delta u_n, \quad \text{(D21)}$$

where $\bar{\alpha}_{pq} = (\alpha_p + \alpha_q)/2$, $\bar{\rho}_{pq} = (\rho_p + \rho_q)/2$, and $v_{\rm sig} = c_p + c_q + \beta \max(\Delta u_n, 0)$. The total face-averaged pressure entering the flux computation is

$$\langle \mathcal{P} \rangle_{pq}^{\rm tot} = \langle P^* \rangle_{pq} + \langle \Pi \rangle_{pq}. \quad \text{(D22)}$$

D.7 Modified Conservation Laws

With the HLLC+CD10 face-averaged quantities, the conservation laws of Section 4 take the modified forms.

D.1 Momentum

Replacing $\mathcal{P}_{pq}$ in equation (48) with the total face-averaged pressure (D22):

$$m_p \frac{d \langle u \rangle_p}{dt} = -\sum_{q \in \mathbb{N}(p)} \langle \mathcal{P} \rangle_{pq}^{\rm tot} \, \Delta S_{pq}. \quad \text{(D23)}$$

The anti-symmetry $\langle \mathcal{P} \rangle_{pq}^{\rm tot} = \langle \mathcal{P} \rangle_{qp}^{\rm tot}$ together with $\Delta S_{pq} = -\Delta S_{qp}$ ensures pairwise momentum conservation.

D.2 Internal Energy

$$m_p \frac{d\langle e\rangle_p}{dt} = -\sum_{q\in\mathbb{N}(p)} \langle \mathcal{P}\rangle_{pq}^{\rm tot}\, \Delta S_{pq}\cdot\left(\langle u\rangle_{pq}^{\rm rad} - \langle u\rangle_p\right), \quad \text{(D24)}$$

where $\langle u\rangle_{pq}^{\rm rad}$ is the composed face velocity of equation (D13). The $P\,dV$ work is captured through the relative velocity $(\langle u\rangle_{pq}^{\rm rad} - \langle u\rangle_p)$, which measures the face-averaged rate of volume change projected onto the face normal.

D.3 Total Energy

The total energy is $\langle E\rangle_p = \frac{1}{2} m_p |\langle u\rangle_p|^2 + m_p \langle e\rangle_p$, and its evolution:

$$\frac{d\langle E\rangle_p}{dt} = -\sum_{q\in\mathbb{N}(p)} \langle \mathcal{P}\rangle_{pq}^{\rm tot}\, \Delta S_{pq}\cdot\langle u\rangle_{pq}^{\rm rad}, \quad \text{(D25)}$$

which is anti-symmetric between p and q, confirming global energy conservation.

D.8 RK4 Time Integration

The system is advanced with the classical fourth-order Runge–Kutta scheme. Let $\mathbf{U}_p = (\mathbf{x}_p, \langle u\rangle_p, \langle e\rangle_p)$ be the state vector. At each of the four sub-stages $s = 1, \ldots, 4$:

(a) **Tessellation:** Build/update the Laguerre diagram (Appendix Appendix J) from current positions $\mathbf{x}_p$ and power radii ω_p. Compute face normals $\hat{n}_{pq}$, face areas $\|\Delta S_{pq}\|$, face centroids $\mathbf{x}_{f,pq}$, and cell volumes Ω_p.

(b) **Cell-averaged thermodynamics:** $\rho_p = m_p/\Omega_p$, $P_p = (\gamma-1)\rho_p\langle e\rangle_p$, $c_p = \sqrt{\gamma P_p/\rho_p}$.

(c) **Cell-averaged gradient** (eq. D2) and **slope limiter** (eq. D6).

(d) **MUSCL extrapolation** (eqs. D7–D9) → face states $(P_L, u_{L,n}, \rho_L, c_L)$, $(P_R, u_{R,n}, \rho_R, c_R)$.

(e) **HLLC solve** (eqs. D10–D11) → $\langle P^*\rangle_{pq}$, u_n^*.

(f) **CD10 viscosity** (eq. D21) → $\langle\Pi\rangle_{pq}$; accumulate $v_{s,p}$.

(g) **Face velocity composition** (eq. D13) → $\langle u\rangle_{pq}^{\rm rad}$.

(h) **Fluxes** (eqs. D23–D24) → stage derivative $\mathbf{k}_s$.

After the four stages, the RK4 combination yields $\mathbf{U}_p^{n+1}$. The CD10 parameter α_p is updated once per full timestep using the cell-averaged divergence from the last sub-stage. The power radius ω_p is then updated following Section 3.2. The global timestep is

$$\Delta t = C_{\rm cfl}\min_p \min_{q\in\mathbb{N}(p)} \frac{2\|q-p\|}{v_{\rm sig}}, \quad \text{(D26)}$$

with $C_{\rm cfl} = 0.2$ for RK4.

Appendix E Adaptive Two-Tier Face Scheme

E.1 Motivation

The M(n,m) face interpolation of Section 5 is computationally efficient and geometrically natural to the Laguerre tessellation, requiring only a single weighted average per face. However, being a central scheme, it lacks upwind physics and produces weak shock resolution without substantial artificial viscosity. The HLLC+CD10 scheme of Section Appendix D, on the other hand, provides excellent shock resolution through MUSCL reconstruction and the Riemann solver, but at three to four times the computational cost per face—gradient computation, slope limiting, left/right state extrapolation, and the HLLC solve.

In typical astrophysical simulations, however, only a small fraction (∼5–10%) of the computational volume is shocked at any given time. This motivates a *two-tier* strategy that applies the full HLLC pipeline only where needed, while the majority of faces use the inexpensive M(n,m) scheme. The key requirement is that the blending be performed on a *per-face* basis to preserve the symmetry $\mathcal{P}_{pq} = \mathcal{P}_{qp}$ and hence conservation.

E.2 Face Blending Factor

The blending factor for face (p,q) is defined as

$$f_{pq} = \min\left(1,\ \frac{\max(\alpha_p,\alpha_q)}{\alpha_{\max}} + \theta_{pq}\right), \quad \text{(E1)}$$

where α_p is the CD10 viscosity coefficient (Section Appendix D) and $\alpha_{\max}$ is its maximum permitted value. The auxiliary term θ_{pq} captures incipient compression not yet reflected in the time-lagged α:

$$\theta_{pq} = \begin{cases} \theta_0, & \text{if } \langle\nabla\cdot u\rangle_p < -\epsilon \text{ or } \langle\nabla\cdot u\rangle_q < -\epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad \text{(E2)}$$

with $\theta_0 \sim 0.1$–0.3. Since both $\max(\alpha_p,\alpha_q)$ and the divergence condition are symmetric under $p \leftrightarrow q$,

$$f_{pq} = f_{qp}, \quad \text{(E3)}$$

which is essential for conservation.

E.3 Blended Face Pressure and Velocity

The blended face pressure is a convex combination of the M(n,m) and HLLC values:

$$\boxed{\langle\mathcal{P}\rangle_{pq}^{\rm blend} = (1-f_{pq})\,\langle\mathcal{P}\rangle_{pq}^{M(n,m)} + f_{pq}\,\langle P^*\rangle_{pq}^{\rm HLLC}} \quad \text{(E4)}$$

Similarly, the normal velocity is blended as

$$\langle u_n\rangle_{pq}^{\rm blend} = (1-f_{pq})\,\langle u_n\rangle_{pq}^{M(n,m)} + f_{pq}\,u_n^{*\,\rm HLLC}. \quad \text{(E5)}$$

Tangential and rotational velocity components are unchanged, as they use the same M(n,m) formula in both schemes.

E.4 Viscous Pressure Blending

The artificial viscosity is likewise blended between the NS stress tensor form (Section 5, eq. C26) and the CD10 signal-velocity form (eq. D21):

$$\langle \Pi \rangle_{pq}^{\mathrm{blend}} = (1 - f_{pq})\, \Pi_{pq}^{\tau} + f_{pq}\, \langle \Pi \rangle_{pq}^{\mathrm{CD10}}. \tag{E6}$$

E.5 Conservation

Conservation requires that the net force on a pair be anti-symmetric: $\boldsymbol{F}_{pq} = -\boldsymbol{F}_{qp}$. This is guaranteed because:

(i) $f_{pq} = f_{qp}$ by construction (eq. E1);

(ii) both $\langle \mathcal{P} \rangle_{pq}^{M(n,m)}$ and $\langle P^* \rangle_{pq}^{\mathrm{HLLC}}$ satisfy the symmetry $\langle \mathcal{P} \rangle_{pq} = \langle \mathcal{P} \rangle_{qp}$;

(iii) a convex combination of symmetric quantities is symmetric;

(iv) together with $\Delta S_{pq} = -\Delta S_{qp}$, pairwise momentum conservation follows.

Total energy conservation is similarly ensured by the anti-symmetry of ΔS_{pq}.

E.6 Computational Cost

The two-tier strategy proceeds as follows: for each face, if $f_{pq} < \varepsilon_f$ (a small threshold), only the M(n, m) interpolation is evaluated; otherwise the full HLLC pipeline (gradient, limiter, extrapolation, Riemann solve) is computed and the result blended.

For typical cosmological simulations where ∼5–10% of the volume is shocked: approximately 90% of faces incur only the M(n, m) cost (normalized to 1×), while the remaining ∼10% require M(n, m) plus HLLC (∼ 4×). The aggregate cost is ∼ 1.3× that of a pure M(n, m) scheme, compared with 4× for a pure HLLC scheme.

E.7 Algorithm Summary

Per RK4 sub-stage, the two-tier scheme proceeds as:

(a) **Tessellation** → cell geometry (Ω_p, ΔS_{pq}, $\hat{n}_{pq}$).

(b) **Cell thermodynamics:** $\rho_p = m_p/\Omega_p$, $P_p = (\gamma - 1)\rho_p e_p$, c_p.

(c) **All faces:** M(n, m) interpolation → $\langle \mathcal{P} \rangle_{pq}^{M(n,m)}$, $\langle u \rangle_{pq}^{M(n,m)}$.

(d) **All cells:** cell-averaged $\langle \boldsymbol{\nabla u} \rangle_p$ and $\langle \nabla \cdot u \rangle_p$ (needed for both the NS stress tensor and the blending criterion).

(e) **All cells:** NS viscous stress $\langle \boldsymbol{\tau} \rangle_p$ with $\nu_p = \alpha_p\, h_p\, c_p$ (eq. C27).

(f) **Shocked faces** ($f_{pq} > \varepsilon_f$): MUSCL gradient + limiter + HLLC → $\langle P^* \rangle_{pq}^{\mathrm{HLLC}}$, $u_n^{*\,\mathrm{HLLC}}$.

(g) **All faces:** blend $\langle \mathcal{P} \rangle_{pq}^{\mathrm{blend}}$ (eq. E4), $\langle \Pi \rangle_{pq}^{\mathrm{blend}}$ (eq. E6).

(h) **Fluxes** (eqs. D23–D24) → stage derivative $\mathbf{k}_s$.

Appendix F Earlier comparison of five face algorithms

The profiles and L_1 table in this appendix compare SPH, a Voronoi scheme with Monaghan viscosity, the Price signal-velocity viscosity, HLLC with the Cullen–Dehnen switch, and a Navier–Stokes stress plus artificial viscosity. These runs do not use the GFS face velocity, and the transmissive problems do not insert inflow particles.

Appendix G A Campaign of Verifications

We validate the one-dimensional Lagrangian scheme by comparing five methods on a suite of eight standard shock-tube problems with $N = 200$ equal-mass particles and $\gamma = 1.4$ (except for the Noh problem, which uses $\gamma = 5/3$). The five methods are:

1. **SPH**: cubic-spline kernel with $(\alpha, \beta) = (1, 2)$ artificial viscosity.

2. **Voronoi**: $M(3, 4)$ face averaging with the Monaghan-type AV using the $\Delta P/\langle P \rangle$-switch.

3. **Price AV**: signal-velocity–based artificial viscosity ($\alpha = 1.5$, $\beta = 2$) without a Riemann solver.

4. **HLLC+CD10**: HLLC Riemann solver combined with the Cullen–Dehnen viscosity switch ($\alpha_{\max} = 1$, $\ell = 0.05$, $\beta = 1$) and MUSCL reconstruction with minmod limiter (Appendix Appendix M).

5. **NS+AV**: Navier–Stokes viscous stress with adaptive viscosity $\nu = \nu_c\, h\, c_s$ ($\nu_c = 0.1$), an acoustic correction $\Delta P/Z$ ($f_{dp} = 0.25$), and Monaghan-type AV ($\alpha = 0.5$, $\beta = 4$). This combination mirrors the multi-dimensional dissipation framework described in Section C.6, where the $M(n, m)$ face stencil provides implicit upwinding and the NS stress supplies physical viscosity. In one dimension the $\Delta P/Z$ acoustic correction serves as the analogue of the $M(n, m)$ transverse upwinding.

All methods employ the fourth-order Runge–Kutta integrator with $C_{\mathrm{cfl}} = 0.2$. Figures G1–G8 present the density, velocity, pressure, and internal-energy profiles for each test problem across the five methods, and Table G1 summarizes the L_1 density errors.

Table G1. $L_1(\rho)$ errors for the 1-D test suite ($N = 200$).

Test	SPH	Voronoi	Price AV	HLLC+CD10	NS+AV
Sod	8.83e-3	8.34e-3	7.11e-3	**5.57e-3**	7.41e-3
Blast	1.30e-1	**5.63e-2**	7.05e-2	9.11e-2	8.87e-2
Noh	1.08e+0	2.26e-2	2.29e-2	2.18e-2	**2.05e-2**
Shu–Osher	5.14e+0	1.24e+0	2.74e+0	6.93e-1	**6.00e-1**
Lax	6.52e-1	6.40e-3	4.81e-3	**4.65e-3**	5.41e-3
Dbl. Raref.	5.08e-1	3.93e-2	3.78e-2	**2.56e-2**	3.32e-2
Collision	7.20e+0	3.70e-1	2.37e+0	**2.14e-1**	2.44e-1
Contact	1.03e+0	5.23e-3	7.30e-3	5.71e-3	**4.67e-3**

Note. Bold entries indicate the best result for each problem. The NS+AV method achieves the lowest total L_1 error ($\sum = 1.004$) across all eight tests, compared to HLLC+CD10 (1.067), Voronoi (1.796), Price AV (5.20), and SPH (15.6).

G.1 Sod Shock Tube

The Sod shock tube (Sod 1978) is a standard Riemann problem with a contact discontinuity, a rarefaction fan, and a shock wave. The initial conditions on $0 \le x \le 1$ are $(\rho, P, u)_L = (1, 1, 0)$ for $x < 0.5$ and $(\rho, P, u)_R = (0.125, 0.1, 0)$ for $x \ge 0.5$, with reflecting walls. Figure G1 shows that all Lagrangian methods capture the shock and rarefaction well. SPH exhibits a visible density undershoot at the contact discontinuity. The HLLC+CD10 method yields the lowest L_1 error (Table G1), followed closely by NS+AV (7.41×10^{-3}) and Price AV (7.11×10^{-3}).

G.2 Interacting Blast Waves

The Woodward–Colella interacting blast-wave problem places two strong shocks in $0 \le x \le 1$ with reflecting walls and the initial pressure distribution $(P_L, P_M, P_R) = (1000, 0.01, 100)$, $\rho = 1$, and $u = 0$. The extreme pressure ratio ($\sim 10^5$) makes this a severe test for any Lagrangian method. Figure G2 shows that the Voronoi method best resolves the interacting shocks, while SPH significantly smears the density peaks. The HLLC+CD10 result overshoots the peak near $x \approx 0.76$, reflecting the sensitivity of the MUSCL reconstruction to the extreme pressure gradient. NS+AV ($L_1 = 8.87 \times 10^{-2}$) performs comparably to HLLC+CD10 (9.11×10^{-2}), capturing the density peaks without the overshoot seen in the Riemann-solver approach.

G.3 Noh Strong Shock

The Noh problem features a cold gas ($e = 10^{-6}$) converging at $u = -1$ toward a reflecting wall at $x = 0$ with $\gamma = 5/3$. The exact solution is an outward-propagating strong shock at $x_s = t/3$ behind which $\rho = 4$ (the strong-shock compression ratio for $\gamma = 5/3$). Figure G3 shows that all Lagrangian methods except SPH achieve $L_1(\rho) \approx 0.02$, accurately capturing the post-shock plateau at $\rho = 4$ with only mild wall-heating at the reflecting boundary. NS+AV achieves the best result ($L_1 = 2.05 \times 10^{-2}$), slightly ahead of HLLC+CD10 (2.18×10^{-2}). SPH is notably worse ($L_1 = 1.08$) due to its kernel-smoothed density estimate, which broadens the shock front.

G.4 Shu–Osher Shock-Entropy Interaction

The Shu–Osher problem propagates a Mach-3 shock into a sinusoidally perturbed density field ($\rho = 1 + 0.2 \sin 5x$ for $x > -4$) on $-5 \le x \le 5$ with transmissive boundaries. The interaction generates high-frequency entropy waves behind the shock, testing a method's ability to preserve small-scale structure. Figure G4 shows that SPH severely damps the post-shock oscillations. NS+AV best preserves the entropy waves ($L_1 = 6.00 \times 10^{-1}$), surpassing HLLC+CD10 (6.93×10^{-1}). The NS viscous stress provides less dissipation in smooth oscillatory regions than the Cullen–Dehnen switch, allowing finer structure to survive. Voronoi (1.24) and Price AV (2.74) suffer from excessive diffusion of the post-shock density peaks at this resolution. These transmissive runs do not insert post-shock gas, so the shock remains near $x \simeq 1.8$. The GFS result with inflow is Figure 3.

G.5 Lax Shock Tube

The Lax problem (Lax & Wendroff 1960) is a Riemann problem with non-zero initial velocity on the left state: $(\rho, P, u)_L = (0.445, 3.528, 0.698)$ and $(\rho, P, u)_R = (0.5, 0.571, 0)$ for $x \ge 0.5$, with transmissive boundaries and $t_{\rm end} = 0.13$. This test produces a strong shock, a contact discontinuity, and a rarefaction fan, similar to the Sod problem but with a faster-moving contact wave. Figure G5 shows that the four non-SPH Lagrangian methods produce excellent results ($L_1 \approx 5 \times 10^{-3}$), with NS+AV ($5.41 \times 10^{-3}$) competitive with the others. SPH fails to resolve the contact discontinuity ($L_1 = 6.52 \times 10^{-1}$).

G.6 Double Rarefaction

The double rarefaction (Toro test #2) initializes two symmetric rarefaction waves diverging from $x = 0.5$: $(\rho, P, u)_L = (1, 0.4, -2)$ and $(\rho, P, u)_R = (1, 0.4, +2)$, with transmissive boundaries and $t_{\rm end} = 0.15$. This problem creates a near-vacuum region at the center where the density drops to $\rho \approx 0.02$ and the pressure approaches zero. Figure G6 shows that the Lagrangian methods accurately resolve the double rarefaction structure, with HLLC+CD10 achieving $L_1 = 2.56 \times 10^{-2}$, followed by NS+AV ($3.32 \times 10^{-2}$) and Voronoi and Price AV at $L_1 \approx 3.8$–3.9×10^{-2}. SPH performs significantly worse ($L_1 = 5.08 \times 10^{-1}$) due to difficulty resolving the near-vacuum region.

G.7 Two-Shock Collision

The two-shock collision (Toro test #4) produces two strong shocks colliding near $x = 0.4$:

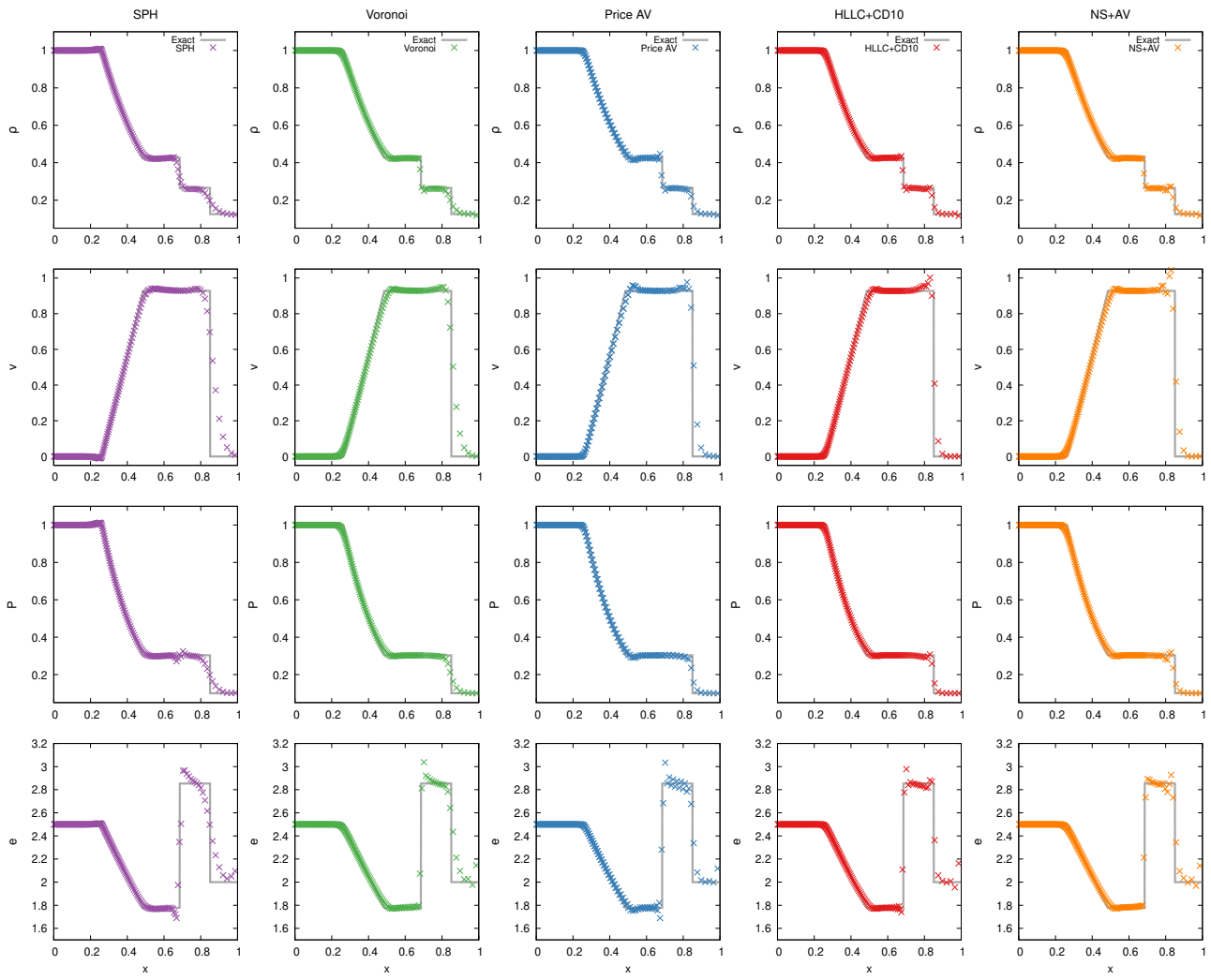

Figure G1. Sod shock tube ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

$(\rho, P, u)_L = (5.99924, 460.894, 19.5975)$ and $(\rho, P, u)_R = (5.99242, 46.095, -6.19633)$, with transmissive boundaries and $t_{\rm end} = 0.035$. The solution contains two shocks, a contact discontinuity, and a very high density peak ($\rho \approx 31$). Figure G7 shows that HLLC+CD10 best resolves the density peak ($L_1 = 2.14 \times 10^{-1}$), with NS+AV close behind (2.44×10^{-1}) and Voronoi at 3.70×10^{-1}. Price AV shows larger errors ($L_1 = 2.37$) from excessive post-shock oscillations, and SPH performs worst ($L_1 = 7.20$). Both boundaries are inflows. Insertion of the exterior state at $N = 200$ reduces the density L_1 from 0.47 to 0.078. The energy error of that calculation is 1.6×10^{-5}.

G.8 Stationary Contact Discontinuity

The stationary contact test isolates the contact discontinuity by imposing equal pressures on both sides: $(\rho, P, u)_L = (1, 1, 0)$ and $(\rho, P, u)_R = (0.125, 1, 0)$ for $x \geq 0.5$, with transmissive boundaries and $t_{\rm end} = 0.25$. The exact solution is a stationary density jump at $x = 0.5$ with zero velocity and uniform pressure everywhere. Figure G8 shows that the NS+AV method achieves the sharpest contact resolution ($L_1 = 4.67 \times 10^{-3}$), closely followed by Voronoi (5.23×10^{-3}), HLLC+CD10 (5.71×10^{-3}), and Price AV (7.30×10^{-3}). The NS viscous stress, being proportional to $\partial v/\partial x$, vanishes identically at a stationary contact where $v = 0$ everywhere, providing no spurious diffusion of the density jump. SPH completely smears the contact ($L_1 = 1.03$), illustrating the well-known zeroth-order inconsistency of the kernel density estimate at sharp discontinuities.

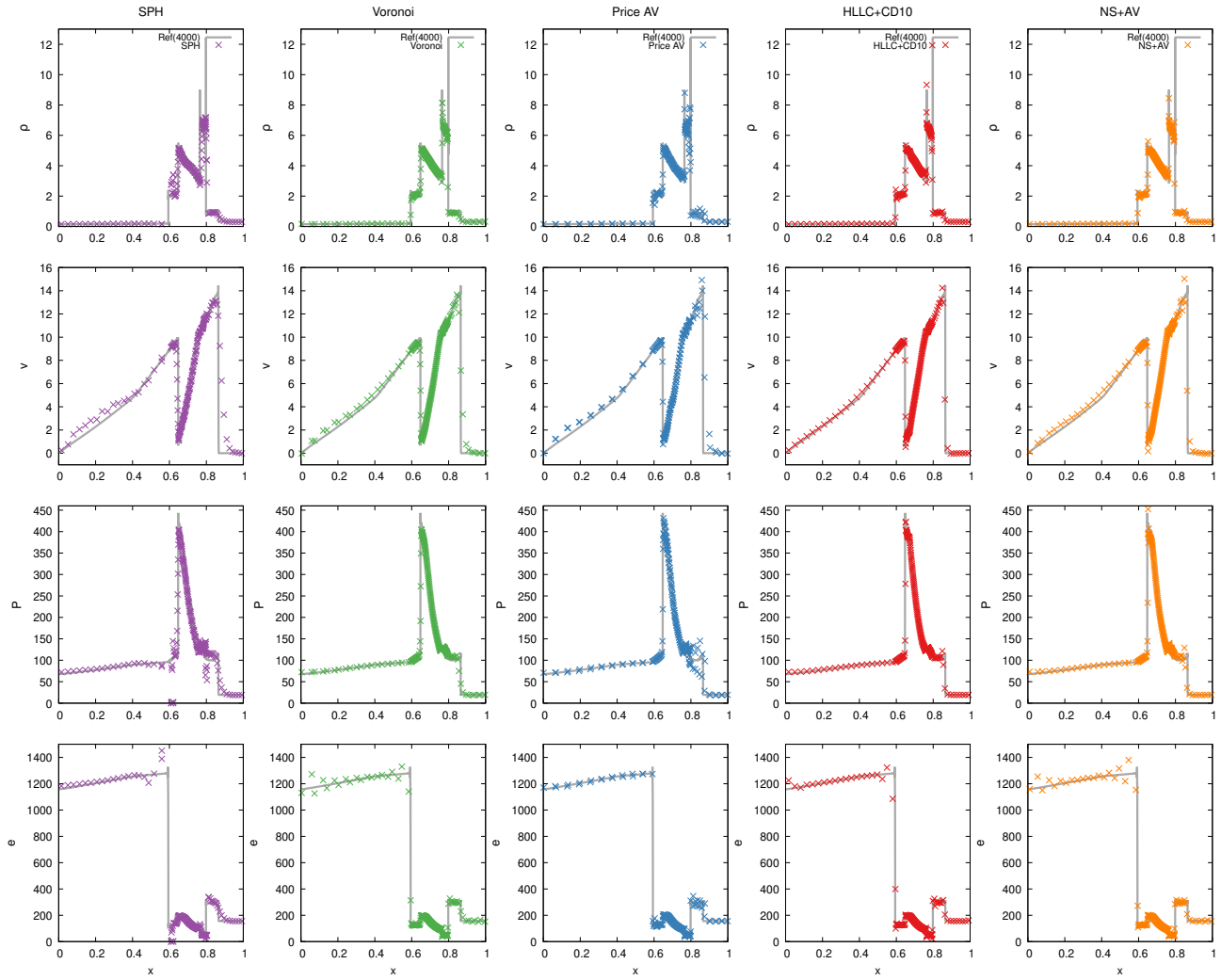

Figure G2. Interacting blast waves ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: high-resolution reference ($N = 4000$); crosses: simulation results.

Appendix H Conservation of Energy

The energy conservation can be obtained by multiplying equation (11) with the flow velocity, u, as

$$u \cdot \boldsymbol{f}_{\rm ext} = u \cdot \frac{\partial}{\partial t}(\rho u) + u \cdot \{\nabla \cdot (\rho u u)\} \tag{H1}$$

$$= \frac{\partial}{\partial t}(\rho u \cdot u) + \nabla \cdot \{\rho u (u \cdot u)\} - \rho u \cdot \left[\frac{\partial u}{\partial t} - (u \cdot \nabla) \cdot u\right]$$

$$= \frac{\partial}{\partial t}(\rho u \cdot u) + \nabla \cdot \{\rho u (u \cdot u)\} - u \cdot \boldsymbol{f}_{\rm ext}, \tag{H2}$$

which leads to

$$\frac{\partial}{\partial t}(\rho k) + \nabla \cdot (\rho k u) = u \cdot \boldsymbol{f}_{\rm ext} \tag{H3}$$

where the kinetic energy of the flow is defined as $k \equiv (u \cdot u)/2$.

Appendix I Barycenter Interpolation

In this section, we briefly describe the interpolation of polytrope in N dimensional space. Consider that there is n-sided polygon and assume the vertex of the polygon to be nodes to interpolate. Then, any point inside the polygon can be written by the affine combination of the nodes $(v_1, v_2, \cdots, v_n)$ as

$$v = \frac{\lambda_1 v_1 + \lambda_2 v_2 \cdots + \lambda_n v_n}{\lambda_1 + \lambda_2 + \cdots + \lambda_n} = w_1 v_1 + w_2 v_2 + \cdots + w_n v_n \tag{I1}$$

where $w_i \equiv \lambda_i / \Sigma_j \lambda_j$.

$$\sum_{i=1}^{n} \lambda_i(v)(v_i - v) = 0 \tag{I2}$$

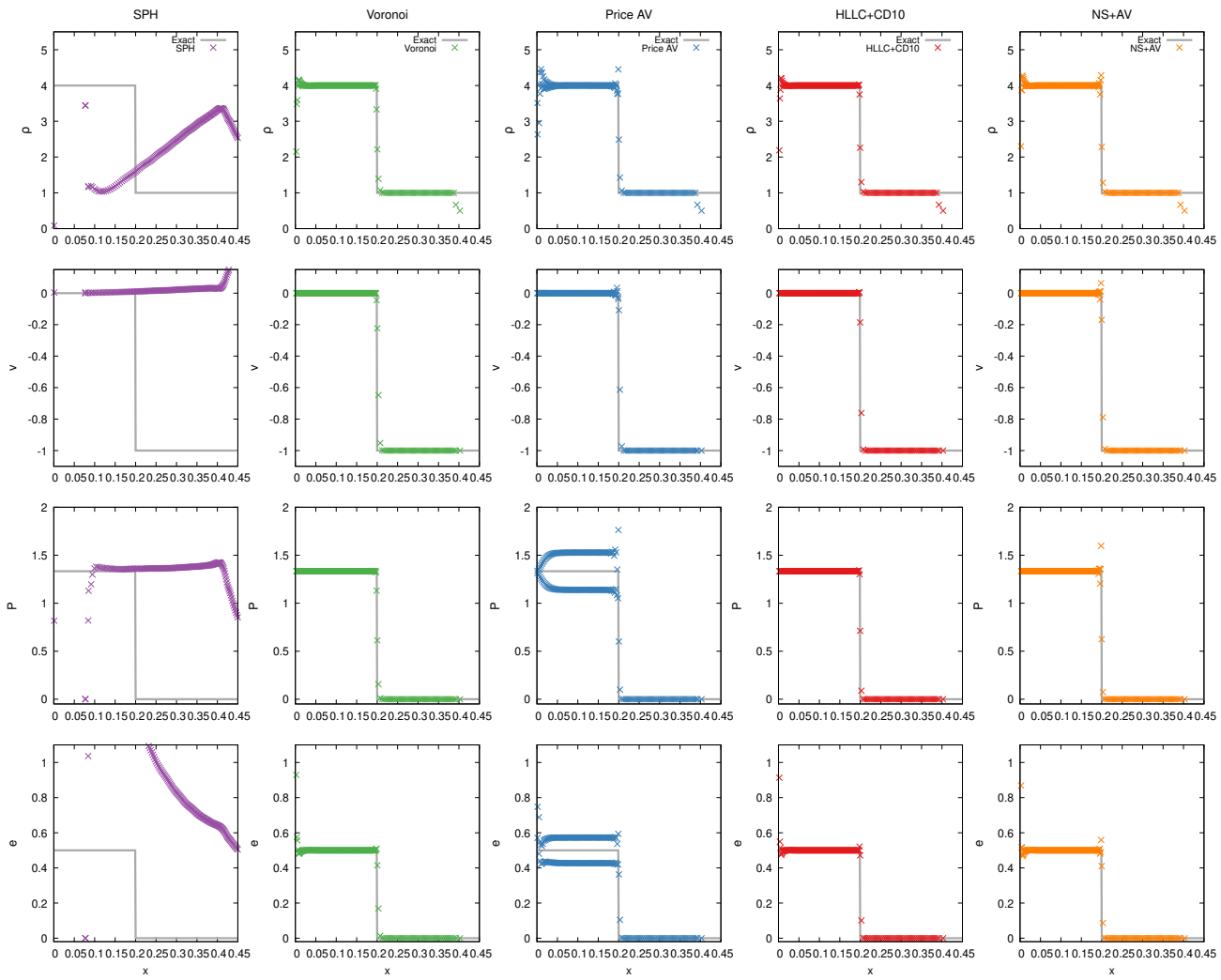

Figure G3. Noh strong shock ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

Appendix J The Recursive Bisection To Generate Laguerre Diagram

For a given set of sites (or particles), we may generate the Laguerre cells. A Laguerre cell is defined as a set of positions which is closer to the site than any other site. We employ a recursive bisection method (RBM) to construct the Voronoi cell of a given site. This spatial partitioning technique determines the volume where all interior positions are closest to the target site. The method can be summarized as follows: we first sort neighboring sites by their distance to the target site. Next, we place a large initial cube centered on the target. From the nearest neighboring site, we draw a bisectional plane (or line in 2D) between the neighbor and the target site, cutting away the outer volume. This process is iterated until the maximum distance of Voronoi-cell vertices from the center is less than twice the next neighboring distance, indicating that further bisection is no longer meaningful in constructing Voronoi cells.

Figure J1 illustrates how the BPM is working in the three dimensions. We initially set a big cubic Voronoi cell centered on the target site (top-left panel) and prune the outer parts starting from the nearest neighbors.

An example of generating the 2D Laguerre cells is shown in Figure J2. The black dots are the semi-randomly distributed particles and their corresponding Laguerre cells are painted in various colors. Here, the semi-random means that we place particle at the regular grid points and randomly perturb their positions within the grid cells.

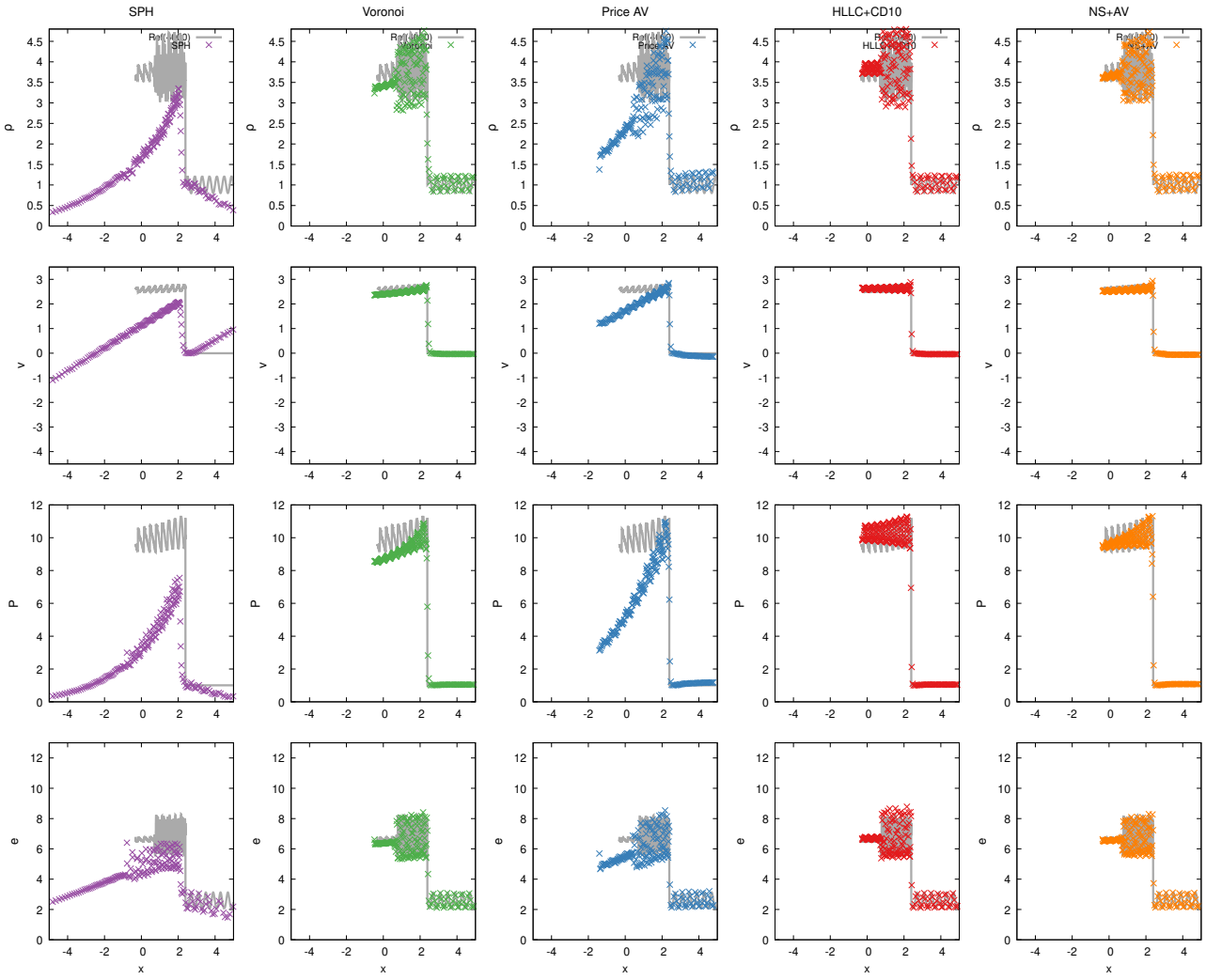

Figure G4. Shu–Osher shock-entropy interaction ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: high-resolution reference ($N = 4000$); crosses: simulation results.

Appendix K Laguerre Face Velocity

K.1 Definitions and Setup

Let two generator sites be defined by their positions p, q and velocities u_p and u_q. Each site has an associated weight w_p, w_q with time derivatives dw_p/dt and dw_q/dt.

We define the following geometric quantities:

- Unit normal vector (pointing from p to q): $n = \frac{q-p}{\|q-p\|}$
- Intersection point (where the radical plane intersects the line connecting p and q):

$$m = \frac{1}{2}\left(p + q + \frac{w_p^2 - w_q^2}{\|q-p\|}n\right) \tag{K1}$$

The Radical Plane is the locus of points x satisfying,

$$\|x-p\|^2 - w_p^2 = \|x-q\|^2 - w_q^2 \tag{K2}$$

from which we may derive the defining equation as,

$$2x\cdot(q-p) = \|q\|^2 - \|p\|^2 - w_q^2 + w_p^2. \tag{K3}$$

Differentiating both sides with respect to time t:

$$\frac{dx}{dt}\cdot(q-p) + x\cdot(u_q-u_p) = q\cdot u_q - p\cdot u_p - w_q\frac{dw_q}{dt} + w_p\frac{dw_p}{dt} \tag{K4}$$

Let $\frac{dx}{dt} = u_{\text{face}}$. We are interested in the normal component $u_n = u_{\text{face}}\cdot n$. Dividing by d:

$$u_n + x\cdot\frac{(u_q-u_p)}{\|q-p\|} = \frac{q\cdot u_q - p\cdot u_p + (w_p\frac{dw_p}{dt} - w_q\frac{dw_q}{dt})}{\|q-p\|} \tag{K5}$$

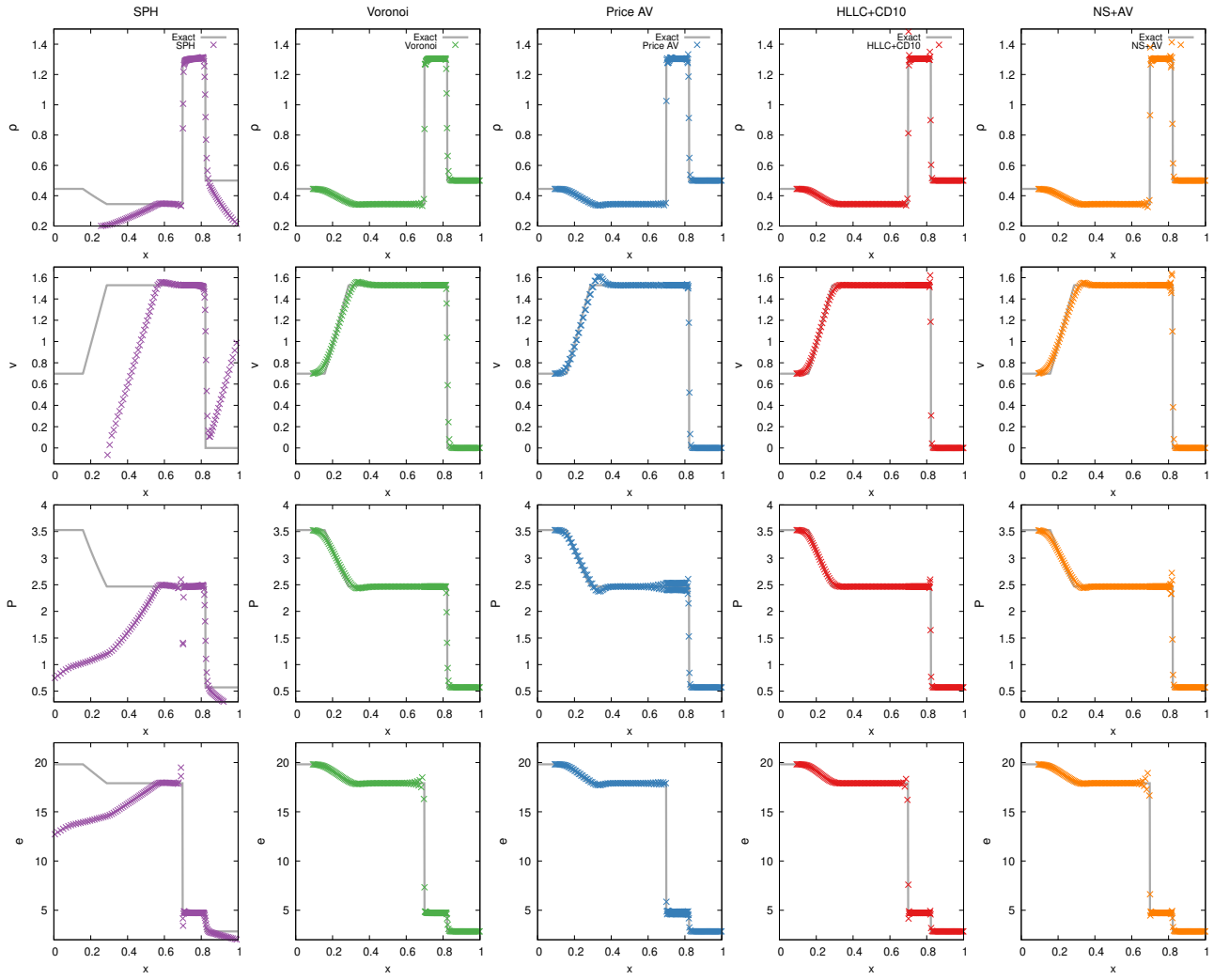

Figure G5. Lax shock tube ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

Solving for u_n and rearranging terms using dot products relative to the particle positions:

$$u_n(x) = \frac{1}{\|q-p\|}\left[u_p \cdot (x-p) - u_q \cdot (x-q) + \left(w_p \frac{dw_p}{dt} - w_q \frac{dw_q}{dt}\right)\right] \tag{K6}$$

K.2 Derivation of Tangential and Rotational Velocity

Geometrically, the tangential velocity of an infinite plane is undefined. However, in the context of dynamic tessellations (like Voronoi/Laguerre moving meshes), the interface is assumed to move rigidly attached to the motion of the defining particles.

The total 3D velocity $u_{\text{face}}(x)$ is the sum of three components:

1. Normal Expansion: The speed along the normal vector derived above.
2. Transverse Translation: The average tangential velocity of the two particles.
3. Rotation (Tilting): The rotation of the plane caused by the relative motion of particles perpendicular to the connecting axis.

K.3 Average Tangential Velocity ($\bar{u}_\perp$)

We extract the tangential components of the particle velocities:

$$u_{p,\perp} = u_p - (u_p \cdot n)n \tag{K7}$$

$$u_{q,\perp} = u_q - (u_q \cdot n)n \tag{K8}$$

The plane translates tangentially with the average of these velocities:

$$\bar{u}_\perp = \frac{u_{p,\perp} + u_{q,\perp}}{2} \tag{K9}$$

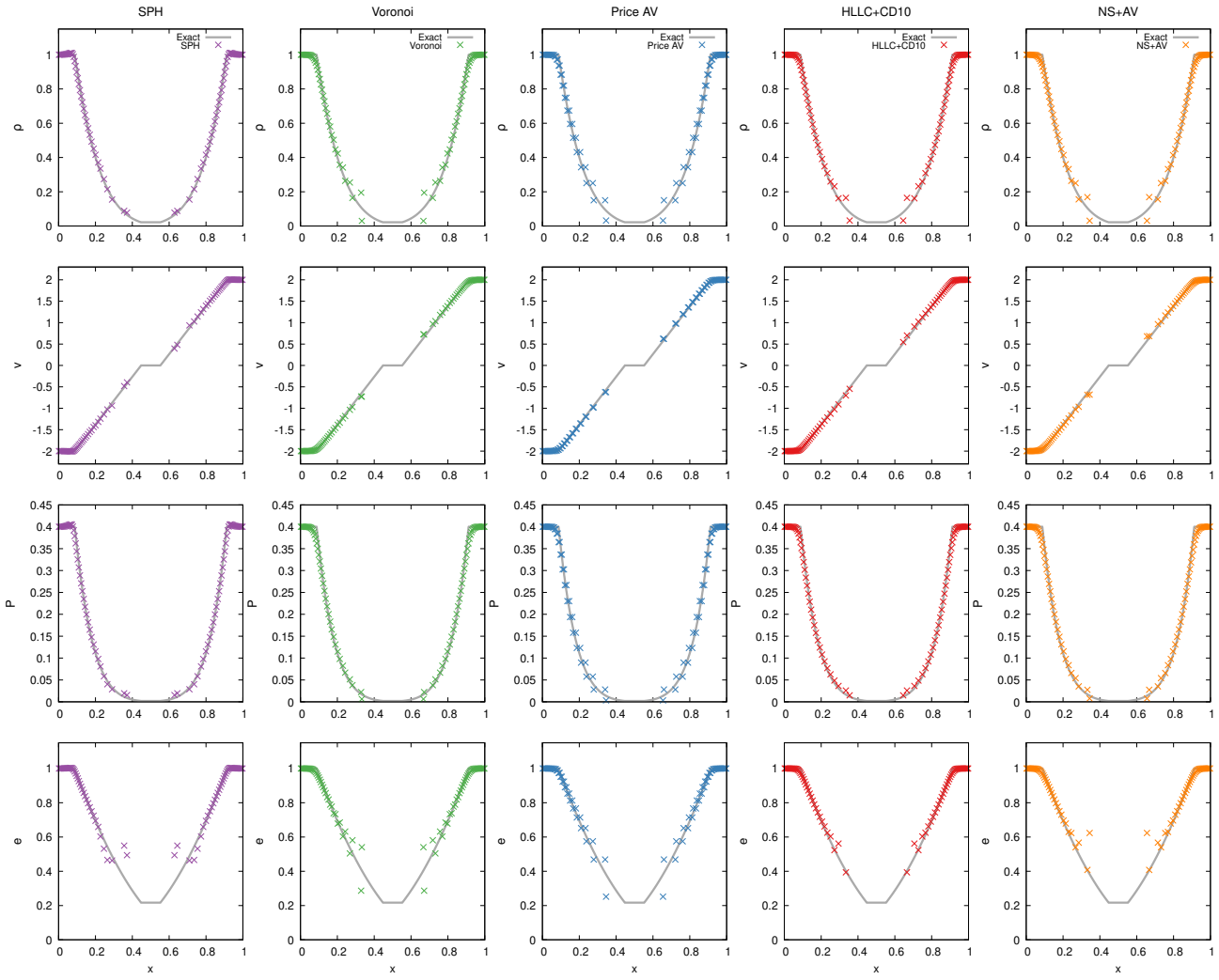

Figure G6. Double rarefaction ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

K.4 Rotational Velocity (u_{rot})

If $u_p \neq u_q$, the axis connecting the particles rotates. The angular velocity $\boldsymbol{\omega}$ of the axis d_{pq} is given by the cross product of the normal and the relative velocity:

$$\boldsymbol{\omega} = \frac{n \times (u_q - u_p)}{d} \tag{K10}$$

The plane rotates with this angular velocity around the anchor point m. The linear velocity due to rotation at a point $\mathbf{x}$ is:

$$u_{\text{rot}}(x) = \boldsymbol{\omega} \times (x - m) \tag{K11}$$

Appendix L Final Formula: The 3D Velocity Vector

Combining all components, the full 3-dimensional velocity vector $u_{\text{face}}(x)$ at any position x on the radical plane is:

$$u_{\text{face}}(x) = \underbrace{u_n(x)n}_{\text{Normal}} + \underbrace{\frac{u_{p,\perp} + u_{q,\perp}}{2}}_{\text{Tangential Translation}} + \underbrace{\left[\frac{n \times (u_q - u_p)}{d}\right] \times (x - m)}_{\text{Rotation (Shear)}} \tag{L1}$$

Expanding the terms for computational implementation:

$$\boxed{\begin{aligned} u_{\text{face}}(x) = &\left[\frac{u_p \cdot (x - p) - u_q \cdot (x - q) + \frac{1}{2}\frac{d}{dt}(w_p^2 - w_q^2)}{d}\right] n \\ &+ \frac{1}{2}\left[(u_p + u_q) - \left((u_p + u_q) \cdot n\right) n\right] \\ &+ \frac{1}{d}\left[n \times (u_q - u_p)\right] \times (x - m) \end{aligned}} \tag{L2}$$

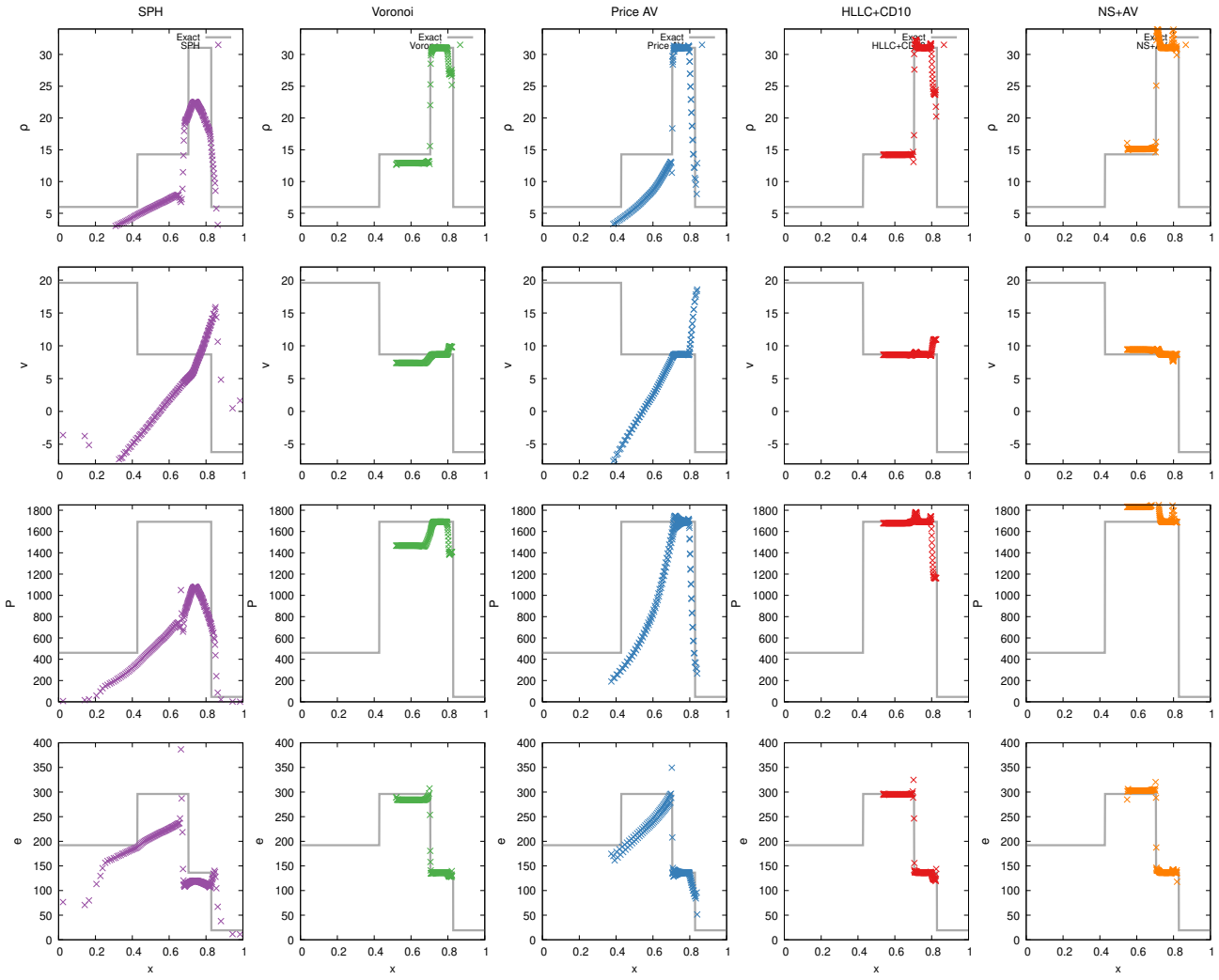

Figure G7. Two-shock collision ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

Appendix M The HLLC + Cullen–Dehnen Algorithm

This appendix describes the complete HLLC + Cullen–Dehnen (CD10) algorithm used as the baseline Laguerre scheme in the 1-D test suite. The method combines an HLLC Riemann solver with a time-dependent artificial-viscosity switch and MUSCL reconstruction, integrated in time with a classical fourth-order Runge–Kutta (RK4) scheme.

M.1 Cell Geometry and Conservation

Each Lagrangian cell p carries a fixed mass m_p. In 1-D, the cell volume (length) is $V_p = x_{p+\frac{1}{2}} - x_{p-\frac{1}{2}}$, where $x_{p+\frac{1}{2}}$ is the face position between cells p and $p+1$. For Laguerre tessellation the face is shifted from the Voronoi midpoint by the weight difference:

$$x_{p+\frac{1}{2}} = \frac{x_p + x_{p+1}}{2} + \frac{\omega_p^2 - \omega_{p+1}^2}{2\, d_{p,p+1}}, \tag{M1}$$

where $d_{p,p+1} = x_{p+1} - x_p$ and ω_p is the Laguerre weight (power radius). When $\omega = 0$ for all cells, the tessellation reduces to the standard Voronoi diagram. The density is updated from $\rho_p = m_p/V_p$ and the thermodynamic state follows from the ideal-gas equation of state, $P_p = (\gamma - 1)\rho_p e_p$ and $c_p = \sqrt{\gamma P_p/\rho_p}$.

M.2 MUSCL Reconstruction

Before solving the Riemann problem at each face, we reconstruct the left and right states using the MUSCL approach with the minmod limiter. For a generic variable ϕ at cell i,

$$\delta\phi_i = \mathrm{minmod}(\phi_i - \phi_{i-1},\ \phi_{i+1} - \phi_i), \tag{M2}$$

where

$$\mathrm{minmod}(a,b) = \begin{cases} \mathrm{sgn}(a)\ \min(|a|,|b|), & ab > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{M3}$$

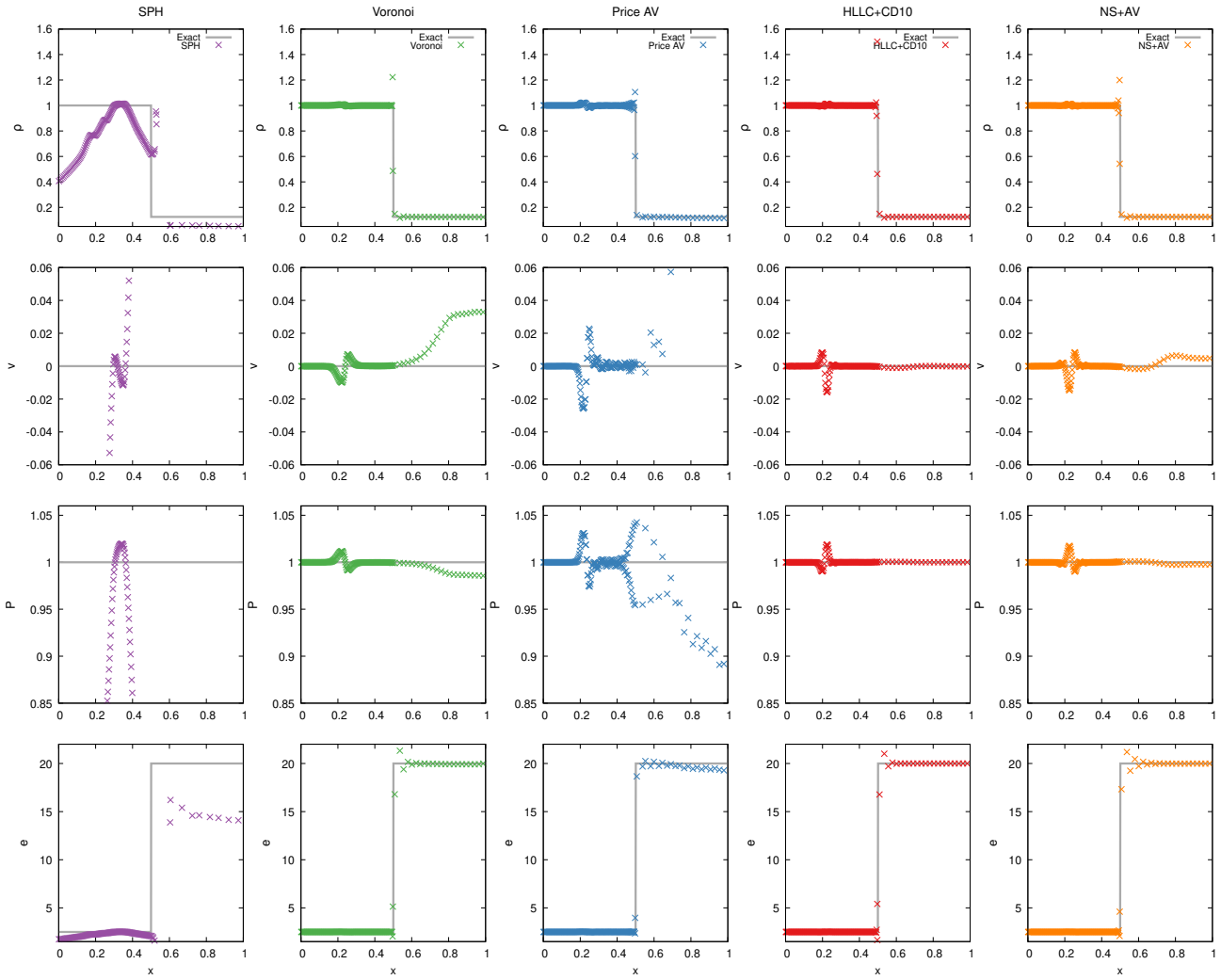

Figure G8. Stationary contact discontinuity ($N = 200$). Columns: SPH, Voronoi, Price AV, HLLC+CD10, NS+AV. Rows: density, velocity, pressure, internal energy. Gray lines: exact solution; crosses: simulation results.

The face states are then extrapolated as

$$\phi_L = \phi_{i-1} + \frac{1}{2}\,\delta\phi_{i-1}, \qquad \phi_R = \phi_i - \frac{1}{2}\,\delta\phi_i. \tag{M4}$$

This reconstruction is applied to the pressure P and velocity u at each internal face.

M.3 HLLC Riemann Solver

Given left/right states (P_L, u_L, ρ_L, c_L) and (P_R, u_R, ρ_R, c_R), the HLLC solver (e.g., Toro 2009) first computes a preliminary pressure estimate via the linearized (PVRS) solver,

$$P_{\rm pvrs} = \frac{Z_R P_L + Z_L P_R + Z_L Z_R (u_L - u_R)}{Z_L + Z_R}, \tag{M5}$$

where $Z_{L,R} = \rho_{L,R}\, c_{L,R}$ are the acoustic impedances. A shock correction factor is then applied to each side:

$$q_K = \begin{cases} \sqrt{1 + \dfrac{\gamma+1}{2\gamma}\left(\dfrac{P_{\rm pvrs}}{P_K} - 1\right)}, & P_{\rm pvrs} > P_K, \\ 1, & \text{otherwise}, \end{cases} \tag{M6}$$

for $K \in \{L, R\}$. The weighted impedances $W_K = \rho_K c_K q_K$ yield the star-state pressure and contact velocity:

$$P^* = \frac{W_R P_L + W_L P_R + W_L W_R (u_L - u_R)}{W_L + W_R}, \tag{M7}$$

$$u^* = \frac{W_L u_L + W_R u_R + P_L - P_R}{W_L + W_R}. \tag{M8}$$

The face pressure is set to $P_{pq} = P^*$ and the face velocity to $u_{pq} = u^*$.

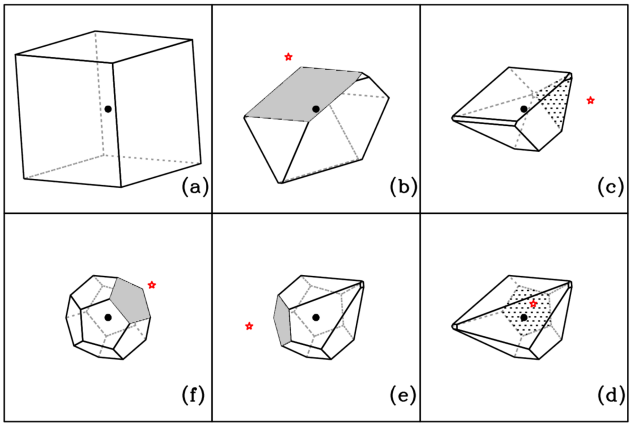

Figure J1. Illustration of the three-dimensional Voronoi cell construction using the BPM. Clockwise from the top left panel we show the update of the Voronoi cell with applying neighbors (red star symbols) one by one. The black circle is the target site. In this figure, we show only a few pruning snapshots (initial shape, 5'th, 8'th, 10'th, 11'th, and 12'th neighbors). To shape this regular dodecahedron we use twelve vertices of the regular icosahedron as neighboring points in the BPM

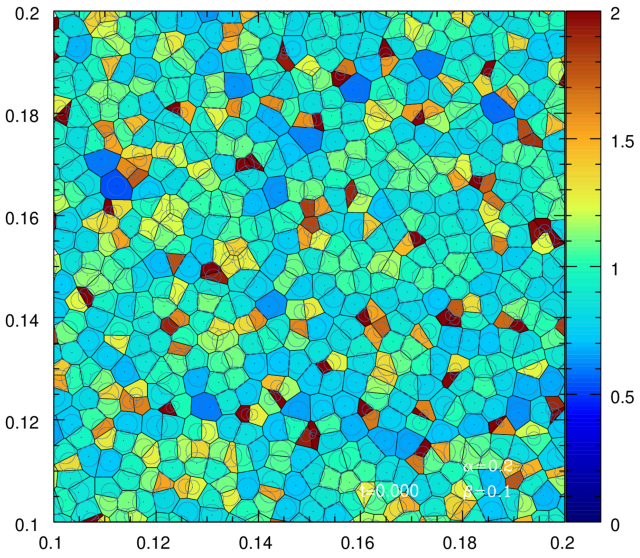

Figure J2. Illustration of the two-dimensional Laguerre diagram. We semi-randomly distribute particles (shown in dots) of equal mass with their w values shown with pale-blue circles. We adopt $\varkappa = 0.55$. Each diagram is color-coded to show the density. The density is normalized to the average.

M.4 Cullen–Dehnen Viscosity Switch

Following Cullen & Dehnen (2010b), a per-cell viscosity parameter α_p is evolved in time. The local activation is driven by the time derivative of the velocity divergence:

$$A_p = \max\left(0,\ -\frac{\nabla\cdot u_p^{(n)} - \nabla\cdot u_p^{(n-1)}}{\Delta t}\right), \tag{M9}$$

where the divergence is computed as $\nabla\cdot u_p = (u_{p+\frac{1}{2}} - u_{p-\frac{1}{2}})/V_p$. The target viscosity coefficient is

$$\alpha_{\rm loc} = \alpha_{\rm max}\,\frac{h_p^2 A_p}{v_{s,p}^2 + h_p^2 A_p}, \tag{M10}$$

where $h_p = V_p$ is the cell size and $v_{s,p}$ is the maximum signal velocity seen by cell p during the current RK4 step. The evolution follows an asymmetric rule:

$$\alpha_p^{(n+1)} = \begin{cases} \alpha_{\rm loc}, & \alpha_{\rm loc} \geq \alpha_p^{(n)}, \\ \alpha_{\rm loc} + (\alpha_p^{(n)} - \alpha_{\rm loc})\, e^{-\Delta t/\tau_p}, & \text{otherwise}, \end{cases} \tag{M11}$$

with the decay timescale $\tau_p = h_p/(2\ell\, v_{s,p})$ and ℓ a dimensionless parameter controlling the decay rate. The coefficient is clamped to $[\alpha_{\rm min},\ \alpha_{\rm max}]$.

The face viscosity pressure is applied only in compression ($\Delta u = u_L - u_R > 0$):

$$\Pi_{pq} = \tfrac{1}{2}\,\bar\alpha_{pq}\, v_{\rm sig}\,\bar\rho_{pq}\,\Delta u, \tag{M12}$$

where $\bar\alpha_{pq} = (\alpha_p + \alpha_q)/2$, $\bar\rho_{pq} = (\rho_p + \rho_q)/2$, and the signal velocity is

$$v_{\rm sig} = c_p + c_q + \beta\,\max(\Delta u,\ 0). \tag{M13}$$

The total face pressure is $P_{pq} + \Pi_{pq}$. In the present 1-D tests we use $\alpha_{\rm max} = 1$, $\ell = 0.05$, $\alpha_{\rm min} = 0$, and $\beta = 1$.

M.5 Flux Computation

The semi-discrete equations of motion for each cell are:

$$\frac{du_p}{dt} = \frac{1}{m_p}\left(\mathcal{P}_{p-\frac{1}{2}} - \mathcal{P}_{p+\frac{1}{2}}\right), \tag{M14}$$

$$\frac{de_p}{dt} = \frac{1}{m_p}\left[\mathcal{P}_{p-\frac{1}{2}}\left(u_{p-\frac{1}{2}} - u_p\right) - \mathcal{P}_{p+\frac{1}{2}}\left(u_{p+\frac{1}{2}} - u_p\right)\right], \tag{M15}$$

where $\mathcal{P}_{p\pm\frac{1}{2}} = P^* + \Pi$ is the total face pressure (HLLC star pressure plus artificial viscosity) and $u_{p\pm\frac{1}{2}} = u^*$ is the HLLC contact velocity at each face. The position evolves as $dx_p/dt = u_p$.

M.6 Boundary Conditions

At a reflecting wall the face velocity is set to zero and the wall pressure is computed from the linearized Riemann condition:

$$P_{\rm wall} = P_p \mp \rho_p c_p u_p, \tag{M16}$$

where the sign depends on the wall side ($-$ for the left wall, $+$ for the right). For a transmissive boundary the face state is simply copied from the adjacent cell: $P_{\rm face} = P_p$, $u_{\rm face} = u_p$.

M.7 RK4 Time Integration

The system is advanced with the standard four-stage Runge–Kutta method. Let $\mathbf{U} = (x,\, u,\, e)$ denote the state vector. Then:

$$\begin{aligned}\mathbf{k}_1 &= f(\mathbf{U}^n),\\ \mathbf{k}_2 &= f\left(\mathbf{U}^n + \tfrac{\Delta t}{2}\,\mathbf{k}_1\right),\\ \mathbf{k}_3 &= f\left(\mathbf{U}^n + \tfrac{\Delta t}{2}\,\mathbf{k}_2\right),\\ \mathbf{k}_4 &= f\left(\mathbf{U}^n + \Delta t\,\mathbf{k}_3\right),\end{aligned} \tag{M17}$$

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \tag{M18}$$

The cell geometry (face positions, volumes, densities) is recomputed at every RK4 sub-stage. The CD10 viscosity parameter α_p is updated once per full timestep after the RK4 combination.

The global timestep is set by the CFL condition:

$$\Delta t = C_{\rm cfl}\,\min_p \frac{V_p}{c_p + |u_p|}, \tag{M19}$$

with $C_{\rm cfl} = 0.2$ for RK4.

M.8 Algorithm Summary

For each full timestep Δt:

1. **RK4 sub-stages** ($s = 1, \ldots, 4$):
   (a) Compute cell geometry: face positions $x_{p+\frac{1}{2}}$, volumes V_p, densities ρ_p, pressures P_p, sound speeds c_p.
   (b) MUSCL reconstruction of (P, u) at each face.
   (c) HLLC Riemann solve → (P^*, u^*) at each face.
   (d) Evaluate CD10 viscosity Π_{pq}; accumulate $v_{s,p}$.
   (e) Apply boundary conditions.
   (f) Compute accelerations (eqs. M14–M15) → stage derivative $\mathbf{k}_s$.
2. **RK4 combination**: update $\mathbf{U}^{n+1}$.
3. **CD10 α update** (eqs. M10–M11).
4. **Timestep**: compute Δt^{n+1} from CFL.

Appendix N Shock Capturing with Adaptive Laguerre Weights

In this appendix we document the stabilization methods explored for the adaptive Laguerre weight (ω) in the presence of strong discontinuities. All tests use the 1-D interacting blast-wave problem (Woodward & Colella 1984; $p_L = 1000$, $p_M = 0.01$, $p_R = 100$, $\gamma = 1.4$, reflecting walls) with $N = 200$ equal-mass particles and the HLLC Riemann solver with the Cullen–Dehnen (2010) viscosity switch.

N.1 Weight Function Modes

Four weight prescriptions $\omega^2 = \varkappa_\omega\, d_{\rm mean}\, \psi(p)$ were tested, where the scaling function ψ is:

(i) PRESSURE: $\psi = (P/P_0)^{(\gamma-1)/\gamma}$,

(ii) ENTHALPY: $\psi = (e + P/\rho)/h_0$,

(iii) SPECVOL: $\psi = (1/\rho)/v_0$,

(iv) CSOUND: $\psi = c_s/c_0$.

Here P_0, h_0, v_0, c_0 are reference scales set from the initial conditions (geometric mean of IC pressures, etc.). The SPECVOL mode is the natural choice for equal-mass Lagrangian particles because $V_i = m/\rho_i$, so $\omega \propto 1/\rho$ makes the Laguerre cell volume track the physical specific volume.

N.2 Area-Adjustment (AA) Correction

When the weight ω changes in time, the Laguerre face shifts even without particle motion. This changes the cell volume and therefore the density, but the standard energy equation does not account for the $P\,dV$ work of this weight-induced compression or expansion.

The AA correction restores energy conservation by adding the weight-change contribution to the face velocity:

$$u_{f,\rm AA} = \frac{1}{2\,\Delta x}\left[\frac{d(w_p^2)}{dt} - \frac{d(w_q^2)}{dt} - \frac{(w_p^2 - w_q^2)(u_q - u_p)}{\Delta x}\right], \tag{N1}$$

so that $u_f \to u_f + u_{f,\rm AA}$ in the Riemann problem on each face. The time derivatives are estimated by finite difference:

$$\frac{d(w_p^2)}{dt} \approx \frac{w_p^{2,n} - w_p^{2,n-1}}{\Delta t}. \tag{N2}$$

A limiter $|u_{f,\rm AA}| \le \alpha_{\rm AA}\, c_s$ is applied to prevent overshoots, with $\alpha_{\rm AA} = 1$ as the default.

An analytical variant replaces the finite difference with $d(\omega^2)/dt = -\beta\,\omega^2\,\nabla\cdot u$, where the coefficient β depends on the weight mode: $\beta = \gamma - 1$ (CSOUND), $\beta = 2(\gamma - 1)$ (PRESSURE, ENTHALPY), $\beta = -2$ (SPECVOL).

N.3 Tested Stabilization Methods

N.1 Method 1: Shock-Switch Weight Damping

A compression-based switch detects shocks and reduces ω^2 locally:

$$\omega^2 \;\longleftarrow\; \frac{\omega^2}{1 + \kappa_s\, S^2}, \qquad S \equiv \frac{\max(0, -\nabla\cdot u)\, d_{\rm mean}}{c_s}, \tag{N3}$$

where κ_s is the damping coefficient. The dimensionless shock indicator S measures the compressive velocity divergence in units of the sound crossing time.

N.2 Method 2: Weight Relaxation

Exponential smoothing of ω^2 toward the target value with a time constant $\tau = \tau_r\, d_{\rm mean}/c_s$:

$$\omega^2_{\rm actual} = \omega^2_{\rm old} + \alpha_r\,(\omega^2_{\rm target} - \omega^2_{\rm old}), \qquad \alpha_r = \min\!\left(1,\ \frac{\Delta t}{\tau}\right). \tag{N4}$$

This prevents sudden jumps in ω^2 and starts from $\omega = 0$ at $t = 0$.

N.3 Method 3: Weight Gradient Limiter

An iterative smoother limits the spatial gradient of ω:

$$\left|\omega_i - \omega_j\right| \ \le\ g_{\rm max}\,\Delta x_{ij} \tag{N5}$$

for each neighboring pair (i, j). When violated, both weights are pulled toward their mean by a factor $g_{\rm max}\Delta x/|\omega_i - \omega_j|$. Three passes are applied per timestep.

N.4 Method 4: Pressure-Ratio Threshold

The shock-switch damping is activated only when the local pressure ratio exceeds a threshold:

$$\max\!\left(\frac{P_i}{P_j},\ \frac{P_j}{P_i}\right) > R_p^{\rm crit} \quad \text{for any neighbor } j. \tag{N6}$$

This is designed to leave moderate shocks (e.g. Shu–Osher) unaffected while targeting extreme discontinuities (e.g. interacting blast waves with P-ratio $\sim 10^5$).

N.4 Results on the Blast Problem

Table N1. Blast-wave results for individual methods (*without* AA correction). All configurations use CSOUND weight mode with $\varkappa_\omega = 0.1$–0.3 and face-shift limit 0.2–0.5. *All entries failed* (hit 50 000 step limit).

Method	Configs tested	Result
Shock-switch alone	24	All FAIL
Weight relaxation alone	24	All FAIL
Gradient limiter alone	24	All FAIL
Combinations of above	36	All FAIL

N.5 Multi-Problem Assessment

N.6 Discussion

Two key findings emerge from these experiments:

1. *AA correction is necessary but not sufficient.* Without AA, the weight-induced volume change is not reflected in the energy equation, leading to a positive feedback loop at density discontinuities where ω changes sharply. With AA, energy is conserved and the Shu–Osher problem improves by 12–14%. However, AA alone cannot stabilize the extreme pressure ratio ($\sim 10^5$) of the blast problem.

Table N2. Blast-wave results with AA correction enabled. SPECVOL weight mode, $\alpha_{\rm AA} = 1.0$.

Configuration	$\varkappa_\omega$	κ_s	$L_1(\rho)$	Status
Base ($\omega = 0$)	0	—	9.11e-2	PASS
AA only	0.10	0	—	FAIL
AA only	0.20	0	—	FAIL
AA + $\kappa_s = 5$	0.10	5	9.02e-2	PASS ⋆
AA + $\kappa_s = 15$	0.08	15	**8.58e-2**	PASS ⋆
AA + $\kappa_s = 15$	0.10	15	9.76e-2	PASS
AA + $\kappa_s = 20$	0.10	20	9.62e-2	PASS
AA + $\kappa_s = 50$	0.08	50	8.66e-2	PASS ⋆

Note. ⋆ denotes $L_1 < L_1^{\rm base}$. The best result (boldface) improves upon the unweighted baseline by 5.7%. The face-shift limit is $f_s = 0.2$ for all entries.

2. *Shock-switch damping stabilizes extreme shocks at the cost of moderate-shock accuracy.* The blast instability originates at *contact discontinuities* where the density gradient is large but $\nabla \cdot u \approx 0$. The shock-switch (S-based) damping is only partially effective because it targets compressions rather than contacts. A sufficiently strong damping coefficient ($\kappa_s \gtrsim 5$) is needed to stabilize the blast, but this indiscriminately suppresses the adaptive weight in moderate-shock regions (Shu–Osher), erasing the accuracy gain from the Laguerre tessellation.

The fundamental trade-off is between adaptivity and stability: the SPECVOL weight naturally adjusts cell volumes to track the physical specific volume ($V_i = m/\rho_i$), providing excellent accuracy for smooth flows and moderate shocks, but creating catastrophic face shifts at extreme density jumps. A contact-discontinuity–specific limiter that preserves the weight in smooth regions while suppressing it only at extreme density ratios remains an open problem.

Appendix O Sutherland–Hodgman Tessellation for GPU Acceleration

In this appendix we describe an alternative polygon-clipping algorithm for constructing 2D Laguerre cells and discuss why it is adopted for the GPU implementation while the original incremental method is retained on the CPU.

O.1 Incremental Bisection (CPU)

The in-house Laguerre tessellation implemented in LagEunha follows the recursive bisection method described in Appendix Appendix J. For each generator i with weight w_i, the algorithm

1. sorts neighbouring generators by Laguerre distance $d^2_{\rm rad} = \frac{1}{4}\,d^2(1 + \Delta w/d^2)^2$,
2. initialises a large bounding polygon (box),

Table N3. $L_1(\rho)$ across four test problems for selected configurations (HLLC+CD10, SpecVol weight, AA enabled). The *base* row has $\omega = 0$ (standard Lagrangian).

Configuration	$\varkappa_\omega$	κ_s	Sod	Blast	Shu–Osher	Noh
Base (ω=0)	0	—	5.575e-3	9.11e-2	7.27e-1	6.14e-1
AA only	0.10	0	5.567e-3	FAIL	**6.35e-1**	6.14e-1
AA only	0.20	0	5.569e-3	FAIL	**6.25e-1**	6.14e-1
AA + κ_s=15	0.08	15	5.571e-3	**8.58e-2**	7.79e-1	6.15e-1
AA + κ_s=0.5	0.10	0.5	5.567e-3	FAIL	6.40e-1	6.15e-1
AA + $g_{\max}$=0.5	0.10	0	5.567e-3	FAIL	**6.32e-1**	6.15e-1

Note. Bold entries indicate improvement over the unweighted baseline. AA-only configurations yield 12–14% improvement on Shu–Osher but fail on the blast problem. Adding shock-switch damping ($\kappa_s = 15$) stabilizes the blast but regresses Shu–Osher by ∼7%.

3. for each neighbour j in ascending $d^2_{\rm rad}$ order, locates the two points where the Laguerre bisector of (i, j) intersects the current polygon boundary (*enter* and *exit* points), then replaces the clipped portion with a single new face,
4. terminates when the farthest surviving vertex is closer than the next neighbour's Laguerre distance.

The polygon is stored as a *doubly-linked circular list* of `Voro2D_Corner` nodes, each carrying vertex coordinates, pointers to the upper/lower neighbour, and pointers to adjacent corners. The force loop traverses this linked list directly, accessing face geometry and neighbour data through pointer chains:

$$\texttt{tmp} \rightarrow \texttt{tmp->upperlink} \rightarrow \cdots \rightarrow \texttt{vorocorner}, \tag{O1}$$

so that tessellation and force evaluation share the same data structure with no intermediate conversion.

This design is efficient on the CPU: each neighbour touches only the two boundary vertices it clips, giving $O(N_{\rm neigh})$ work per cell for an already-sorted neighbour list. The linked-list storage allows $O(1)$ insertion and deletion of corners. OpenMP thread safety is achieved by allocating one corner pool per thread (or per cell-row).

O.2 Sutherland–Hodgman Clipping (GPU)

The incremental algorithm relies on three features that are fundamentally incompatible with GPU execution:

- *Pointer-based linked lists*: GPUs achieve throughput through coalesced memory access; pointer chasing across heap-allocated nodes destroys spatial locality.
- *Dynamic memory allocation*: each `CreateNewCorner` call requires `malloc`; GPU `malloc` (device-side) is orders of magnitude slower than host `malloc` and serialises warps.
- *Sequential polygon traversal*: finding the enter/exit points requires walking the linked list from an arbitrary active corner, creating warp divergence.

We therefore adopt the Sutherland–Hodgman (S-H) polygon clipping algorithm (Sutherland & Hodgman 1974) for the GPU kernel. S-H processes each half-plane (bisector) by iterating over all edges of the current polygon and classifying each edge–vertex pair as inside or outside:

1. *Both inside*: output vertex A unchanged.
2. *A inside, B outside*: output A, then the intersection.
3. *A outside, B inside*: output the intersection only.
4. *Both outside*: skip.

The polygon is stored as a *fixed-size array* (`poly_x[]`, `poly_y[]`, `poly_neigh[]`) in GPU registers or local memory, requiring no dynamic allocation. Neighbour data is gathered from a cell-linked-list CSR structure and sorted by $d^2_{\rm rad}$ using an insertion sort (small N).

The Laguerre weight enters through the half-plane threshold. For neighbour j at relative position (n_x, n_y) with $d^2 = n_x^2 + n_y^2$, the weight fraction is

$$f_w = \frac{1}{2} + \frac{w_i^2 - w_j^2}{2\,d^2}, \tag{O2}$$

and a polygon vertex (v_x, v_y) lies inside the half-plane of generator i if $v_x n_x + v_y n_y \le f_w\, d^2$. When $w_i = w_j$ this reduces to the standard Voronoi bisector at the midpoint.

O.1 Degenerate-Face Merging

On a regular or near-regular grid, four (or more) Laguerre cells meet at a single point. The incremental algorithm handles this naturally—the enter and exit points of a bisector coincide, so no new face is created. S-H, however, processes edges independently and produces two nearly-identical vertices, creating a degenerate face of near-zero area. Because the CFL time step satisfies $\Delta t \propto A_{\rm face}/v_{\rm sig}$, such faces cause catastrophic time-step collapse.

We eliminate degenerate faces by a post-clipping vertex-merging pass. After all bisectors have been applied, we scan adjacent vertices and remove any pair separated by less than a relative threshold:

$$|\Delta \mathbf{v}|^2 < \epsilon^2_{\rm merge} \equiv 10^{-6}\,\ell^2_{\max}, \tag{O3}$$

where $\ell_{\max}$ is the longest edge of the current polygon. The threshold 10^{-6} (i.e. edges shorter than 10^{-3} times the longest

edge) was determined empirically by comparing GPU and CPU face counts across $\sim 10^5$ Laguerre cells on a perturbed KH-instability grid (Section O.4).

O.3 Thread Safety and Parallelism

Both algorithms are embarrassingly parallel over generators: each cell's tessellation depends only on the positions and weights of neighbours within a bounded stencil.

CPU (OpenMP).—The tessellation loop is parallelised over cell rows (`iy`). Each thread owns a private corner pool (`Voro2D_Corner vorocorner[mp]`) and allocates per-cell neighbour work arrays. Race conditions are avoided because the output (volume, density, gradients, stress) is written to each particle's own struct fields.

GPU (CUDA).—One CUDA thread processes one generator. The kernel `voronoi_tessellate_kernel` gathers neighbours from a 3×3 cell stencil stored in CSR format, performs S-H clipping, vertex merging, and computes volume, density, pressure, sound speed, and (when `av_mode` ≥ 1) velocity gradients and NS stress—all within a single kernel launch. The resulting polygon is written to a `PolygonOut` buffer (fixed 48 corners per particle), then a second kernel (`scatter_faces_kernel`) converts the polygon vertices into a global `FaceCSR` structure (CSR-formatted face connectivity and geometry) consumed directly by the GPU force kernel.

O.4 Performance Comparison

Table O1 summarises the tessellation cost for $N_x = 256$ (131,072 particles) on a single MPI rank. The S-H algorithm is approximately 13% faster than the incremental method on the CPU, owing to cache-friendly array traversal versus pointer-chasing through the linked list. Memory usage per thread is comparable: ∼7 KB (S-H, stack arrays) versus ∼70 KB (incremental, corner pool), but in both cases the dominant allocation is the shared neighbour array (∼5 KB per cell). The GPU kernel achieves an additional 3–4× speedup over the CPU S-H, and when combined with the GPU force kernel the full-step acceleration reaches 24× for `av_mode = 0`.

Table O1. Tessellation time for the KH test ($N_x = 256$, single rank, single thread for CPU). "Incremental" is the in-house recursive bisection; "S-H" is Sutherland–Hodgman on the same CPU.

Method	Time (ms)	Faces	Speedup
Incremental (CPU)	1085	524 288	1.0×
S-H (CPU)	956	524 288	1.1×
S-H (GPU, RTX 5000 Ada)	270	524 288	4.0×

The S-H method produces identical face counts and total cell volumes to the incremental method on the initial (regular) grid. After particles have moved, the GPU S-H generates ∼ 7% more faces than the CPU incremental method; the extra faces are near-degenerate edges at grid corners that survive the merging threshold (O3). These contribute negligibly to the volume integral and do not affect the CFL time step once the merging threshold is applied.

O.5 Design Rationale

Given the modest 13% CPU speedup and the significant refactoring cost—the force loop (`getAccVoro2DBlend`) is tightly coupled to the linked-list corner traversal (O1), requiring a full rewrite to accept array-based face data—we retain the incremental method for all CPU code paths. The S-H algorithm is used exclusively on the GPU, where the array-based `FaceCSR` format is the natural interface to the force kernel. This hybrid design allows both paths to coexist: the CPU path serves as a reference implementation and fallback, while the GPU path provides the production-level throughput needed for large simulations.

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